Commitment Floors for Tipping-Point Commons: Escaping Nash Traps in Multi-Agent Reinforcement Learning

Anonymous Author(s)

Abstract

When must multi-agent systems move beyond self-interest, and how much commitment is enough for collective survival? We study **Tipping-Point Social Dilemmas** (TPSDs)—multi-agent settings where thresholded resource dynamics can trigger irreversible collapse—and identify an extinction-type failure we term the **Nash Trap**: gradient-based learners converge to locally stable under-commitment equilibria that are individually rational yet collectively catastrophic, *even when cooperation is individually rational* ($M/N \geq 1$).

We make five contributions: **(C1)** We show the Nash Trap is *universal across algorithms*: five families (REINFORCE, PPO, IQL, QMIX, LOLA), nine implementations, and 12 M/N conditions all converge to $\lambda \approx 0.5$ with 100% trap rate (20 seeds each). **(C2)** We prove escape requires $\Omega(e^{cN})$ gradient evaluations (Theorem 3), via signal dilution, basin dominance, and exponential drift analysis—confirmed empirically for $N \in \{5, \ldots, 100\}$. **(C3)** We derive commitment floors from recovery dynamics and show they bypass the barrier in $O(1)$ steps (Corollary 4). **(C4)** We introduce **MACCL** (Multi-Agent Constrained Commitment Learning), a primal-dual algorithm that learns state-dependent floors with convergence guarantees (Theorem 5), improving welfare by 203% over fixed floors while maintaining 100% survival. **(C5)** We validate across three environment classes (PGG, CPR, Cleanup), adversarial conditions (30% Byzantine), and scale (N=100).

Together, these results establish the Nash Trap as a **fundamental computational barrier**—not an artifact of limited capacity—and the commitment floor as a principled, polynomial-time bypass.

1 Introduction

Fisheries collapse when each vessel rationally maximizes its own catch. Climate negotiations stall when each nation rationally free-rides on others' emissions cuts. These are not failures of intent but of *individually rational* decision-making near shared tipping points: once a critical resource crosses a collapse threshold, recovery may be slow or effectively irreversible. The central question is not whether cooperation is *achievable in principle* (the Folk Theorem guarantees its existence), but whether standard learning dynamics *reliably converge* to sufficient commitment for collective survival. We prove they do not—and show how to fix this.

We organize our analysis around three diagnostic layers: (i) when the marginal return from cooperation is below its individual cost ($M/N < 1$), agents face a persistent local anti-commitment incentive; (ii) when this misalignment interacts with tipping-point collapse dynamics, it manifests as an extinction-type Nash Trap; and (iii) the severity of recovery near collapse determines how large the commitment floor must be for escape, with severe regimes saturating at maximal feasible commitment. Building on this analysis, we make five contributions:

(C1) **Nash Trap Analysis**: We define the class of Tipping-Point Social Dilemmas (Definition 2), prove a drift condition on resource dynamics (Lemma 6), and establish the Nash Trap as a locally stable fixed point (Proposition 9). Crucially, we demonstrate that the Nash Trap persists *independently of the M/N ratio*: across 12 conditions from $M/N = 0.05$ to $M/N = 1.0$ with 20 seeds each, gradient learners converge to $\lambda \approx 0.50$ with 100% trap rate (Section 4.6, Table 9). This failure is consistent across five algorithm families (REINFORCE, PPO, IQL, QMIX [25], LOLA [8]; nine implementations) under a unified fair-comparison protocol.

(C2) **Computational Intractability (Theorem 3)**: We prove that escaping the Nash Trap requires $\Omega(e^{cN})$ gradient evaluations, establishing three mechanisms: (i) signal dilution: the survival gradient is $O(\epsilon/N)$ against reward noise $O(1)$; (ii) basin dominance: the trap basin has measure $\geq 1 - e^{-c_1 N}$ under uniform initialization; (iii) exponential escape time from drift analysis.

Empirical verification across $N \in \{5, 10, 20, 50, 100\}$ with 20 seeds confirms 100% trap rate and 0% escape rate (Section 4.5).

(C3) **Commitment Floor as Computational Bypass (Corollary 4)**: The commitment floor transforms an exponentially hard optimization into a constrained problem solvable in $O(1)$ gradient steps by excluding the trap basin from the feasible action space.

(C4) **Multi-Agent Constrained Commitment Learning (MACCL)**: We introduce a primal-dual algorithm with formal convergence guarantees (Theorem 5) that learns the *minimum sufficient* state-dependent floor $\phi_1(R_t; \omega)$. In Cleanup environments, MACCL improves welfare by 203% over fixed ϕ_1=1.0 while maintaining 100% survival (Section 4.8, Table 10); in severe TPSDs, it automatically recovers $\phi_1 \approx 1.0$.

(C5) **Cross-Environment Validation**: We validate the framework across three structurally distinct environment classes—PGG (provision), CPR (appropriation), Cleanup (pollution)—under adversarial conditions (30% Byzantine) and scale (N=100), *plus* Google DeepMind's Melting Pot benchmark (Appendix). The Nash Trap reproduces in all TPSD environments; MACCL generalizes across all three classes.

Together, these results establish that the Nash Trap is a **fundamental computational barrier** inherent to tipping-point dynamics—not merely an artifact of $M/N < 1$ or limited algorithm capacity. The commitment floor provides a principled bypass, and MACCL learns where to set it.

2 Related Work

Social Dilemmas in MARL. The tension between individual and collective rationality has deep roots in evolutionary game theory [21, 29] and has been studied in matrix games [16] and sequential social dilemmas [12, 7]. Intrinsic motivation approaches—inequity aversion [12], social influence [14], prosocial agents [24], reputation mechanisms [4], and norm enforcement [15]—promote cooperation but implicitly encode cooperative preferences, making it impossible to distinguish genuine emergence from training conditioning.

Opponent Shaping. LOLA [8] differentiates through opponent updates to shape learning in 2-player games. M-FOS [19] extends this model-free via meta-learning, and POLA [37] adds proximal regularization. More recently, K-Level Policy Gradients (KPG) [28] pursue recursive opponent anticipation without explicit white-box access, but incur $\mathcal{O}(K \cdot N)$ computational cost per agent per step. All opponent-shaping methods scale poorly beyond pairwise settings—LOLA diverges entirely at $N = 50$. Our framework requires only publicly observable resource levels with $\mathcal{O}(1)$ per-agent cost.

Non-linear Dynamics and Tipping Points. While common-pool resource games have been studied [22] and recent MARL benchmarks have expanded the scope of evaluation [23, 5], most environments use linear or stationary dynamics. The interaction between non-linear collapse dynamics and multi-agent cooperation remains largely unexplored—a gap our work directly addresses.

Relationship to Folk Theorem. Classical Folk Theorems guarantee cooperative equilibria exist in infinitely repeated games with sufficient patience ($\delta \to 1$) (see Remark 1). Our Nash Trap does not contradict this: the Folk Theorem guarantees *existence* but not *discoverability* by gradient-based learners. The Nash Trap is a *computational* phenomenon—gradient ascent on individual returns converges to a locally stable fixed point ($\lambda \approx 0.5$) that is Pareto-dominated. The critical distinction is that our game has *non-stationary* dynamics: the resource state R_t creates history-dependent payoffs absent in standard repeated games, and tipping-point collapse makes the effective horizon finite and stochastic.

Safe RL and Constrained Optimization. The constrained MDP framework [3] underpins methods such as CPO [1], safe exploration benchmarks [27], and risk-sensitive objectives (CVaR) that provide safety guarantees by constraining reward optimization [9]. In the multi-agent setting, MACPO and MAPPO-Lagrangian [10] extend constrained optimization to cooperative MARL, while RCPO [34] transforms constraints into reward penalties. These methods assume the constraint (e.g., collision avoidance) is *exogenous and differentiable*. In contrast, our constraint—the commitment floor—is *derived from the environment's tipping-point structure* and addresses an *endogenous* failure mode (the Nash Trap) that no unconstrained method escapes in our tests. We formulate the commitment problem through the constrained lens: a Lagrangian dual approach independently recovers $\phi_1^* = 1.0$ as the safety-optimal floor (Eq. 24), and MACCL extends this with formal convergence guarantees (Theorem 5). Crucially, while CPO constrains a single agent's cost to remain below a threshold, MACCL constrains all agents' *minimum contribution* via a collective floor—transforming the exponentially hard coordination problem (Theorem 3) into a tractable constrained optimization.

Contract-Based and Norm-Based Approaches. Recent work explores mechanisms where agents propose, negotiate, and accept binding contracts to mitigate social dilemmas [6]. Meta-ranking differs in that cooperation arises from pre-specified moral structure rather than runtime negotiation, requiring no communication channel between agents.

Moral Theory in Reward Design. Tennant et al. [33] systematically embed moral theories (utilitarian, deontological, virtue ethics) into MARL reward functions for social dilemma analysis. Our contribution extends this line by identifying *when* each moral commitment level is sufficient for survival: the Moral Commitment Spectrum provides a diagnostic mapping from environmental severity to the required degree of commitment, classifying crisis-response profiles rather than imposing a monotonic ranking.

3 Method

3.1 Meta-Ranking Reward Function

Definition 1 (Meta-Ranking Agent)**.** *Agent i with SVO angle θ_i maximizes:*

$$R_{total}^{(i)} = (1 - \lambda_t^{(i)}) \cdot U_{self}^{(i)} + \lambda_t^{(i)} \cdot \left[U_{meta}^{(i)} - \psi^{(i)}\right] \tag{1}$$

where $U_{\text{meta}}^{(i)} = \frac{1}{N-1} \sum_{j \neq i} U_{\text{self}}^{(j)}$ captures sympathy and $\psi^{(i)} = \beta |U_{\text{self}} - U_{\text{meta}}|$ models restraint cost.

3.2 Dynamic Commitment Mechanism

The key innovation is that λ_t adapts to resource level R_t:

$$\lambda_t^{(i)} = \alpha \lambda_{t-1}^{(i)} + (1 - \alpha) \cdot g(\theta_i, R_t) \tag{2}$$

where:

$$g(\theta, R) = \begin{cases} \max(0, \sin\theta \cdot 0.3) & R < R_{\text{crisis}} \\ \min(1, 1.5 \sin\theta) & R > R_{\text{abundance}} \\ \sin\theta \cdot (0.7 + 1.6R) & \text{otherwise} \end{cases} \tag{3}$$

This implements Sen's insight computationally: when resources are scarce, agents reduce commitment—trading restraint cost against short-term individual payoff. We show below (Section 4.2) that in TPSDs this heuristic can be structurally misaligned with collective safety, because reducing commitment during a crisis accelerates passage through the tipping point.

3.3 Environment Suite

Linear PGG. N agents each receive endowment E and contribute $c_t^i = E \cdot \lambda_t^i$. Individual payoff: $r_t^i = (E - c_t^i) + M \sum_j c_t^j / N$ with multiplier $M = 1.6$. Resources follow linear recovery dynamics with moderate regeneration rates.

Non-linear PGG with Tipping Points. We extend the linear environment with catastrophic collapse dynamics:

$$R_{t+1} = \text{clip}(R_t + f(R_t) \cdot (\bar{c}_t/E - 0.4) - \xi_t,\ 0,\ 1) \tag{4}$$

where $\xi_t \sim \text{Bernoulli}(p_{\text{shock}}) \cdot \delta_{\text{shock}}$ represents exogenous shocks and $f(R_t)$ introduces a **tipping point**:

$$f(R_t) = \begin{cases} 0.01 & R_t < R_{\text{crit}} \quad \text{(near-irreversible collapse)} \\ 0.03 & R_{\text{crit}} \leq R_t < R_{\text{recov}} \quad \text{(hysteresis)} \\ 0.10 & R_t \geq R_{\text{recov}} \quad \text{(normal recovery)} \end{cases} \tag{5}$$

with $R_{\text{crit}} = 0.15$ and $R_{\text{recov}} = 0.25$. When $R_t \leq 0$, the episode terminates and all future rewards become zero.

4 Experiments

Using the environment (Section 3.3) and agent framework from Section 3, we systematically evaluate cooperation across linear (Section 4.1) and non-linear environments.

Evaluation Protocol and Metric Definitions. All experiments use the following standardized evaluation:

- **Episode horizon**: $T = 50$ steps per episode.
- **Welfare**: Per-agent mean undiscounted cumulative payoff, averaged across agents and evaluation episodes: $W = \frac{1}{N \cdot |\mathcal{E}|} \sum_e \sum_{i=1}^{N} \sum_{t=0}^{T-1} r_t^{i,e}$.
- **Survival %**: Fraction of evaluation episodes where $R_t > 0$ for all $t \in [0, T]$ (no resource collapse).
- **Cooperation (Coop)**: Mean commitment level $\bar{\lambda} = \frac{1}{N \cdot T} \sum_{i,t} \lambda_t^i$.
- **Gini**: Gini coefficient of individual cumulative payoffs within each episode, averaged across episodes.
- **Inv.↓**: Probability that a single selfish mutant (λ=0) achieves higher fitness than the group mean over 100 invasion trials (lower is better).

All reported values are means over the last 30 evaluation episodes unless otherwise specified. Error bars (where shown) indicate ± 1 standard deviation across seeds.

4.1 Situational Commitment in Linear Environments

We first validate the meta-ranking framework across 8 environments ($N = 20$–1000, 10 seeds; see Table 22). This result establishes a strong baseline: Situational Commitment is empirically optimal across all tested linear environment configurations. The subsequent sections demonstrate that this demonstrated optimality **catastrophically collapses** under non-linear tipping points—a finding that would be impossible to articulate without first establishing the linear baseline.

Table 1: Moral theory comparison (linear PGG, N=50, 10 seeds). "Inv.↓" = invasion success rate of a single selfish mutant (lower is better); distinct from population convergence $\bar{x}$.

Theory	Coop	Welfare	Sustain.	Inv.↓
Utilitarian	1.000	147.7	100%	99.6%
Situational (Ours)	1.000	147.7	100%	0.1%
Deontological	1.000	129.7	100%	0.1%
Virtue Ethics	0.684	118.9	100%	0.1%
Selfish	0.000	102.8	26%	0.1%

Table 2: Architectural comparison (linear PGG, N=50, 10 seeds). Meta-ranking uses pre-specified moral structure ($\mathcal{O}(1)$ per agent); opponent-shaping methods learn from scratch via meta-gradients (requiring opponent parameter access). "Byz" = cooperation under 50% Byzantine adversaries.

Method	Coop	Welfare	Gini	Byz	Decentr.
M-FOS	1.000	156.3	0.000	0.143	×
POLA	1.000	148.3	0.000	0.504	×
LOLA	0.000	101.5	0.014	0.000	×
LOPT	0.000	101.4	0.014	0.000	×
Meta-Ranking (Ours)	1.000	148.0	0.011	**0.500**	✓

Moral Theory Comparison. Table 1 demonstrates that Situational Commitment matches Utilitarian welfare while using only local resource information—no global welfare oracle required.

SOTA Comparison. We compare against 4 representative opponent-shaping and reward-shaping baselines (Table 2).

Key finding: M-FOS achieves the highest benign welfare (156.3) via meta-learning opponent shaping, but collapses under Byzantine adversaries (0.143). POLA maintains 0.504 Byzantine cooperation, matching meta-ranking's 0.500, but requires white-box access to opponent parameters—precluding decentralized deployment. Meta-ranking is the only method simultaneously achieving high Byzantine resilience and full decentralization.

Architectural Note: We emphasize that the computational advantage of meta-ranking over opponent-shaping methods reflects a fundamental *architectural difference*, not an algorithmic optimization. Meta-ranking operates through internal moral commitment conditioned on observable resource levels ($\mathcal{O}(1)$ per agent), while LOLA/POLA/M-FOS compute meta-gradients over opponent parameter spaces—a categorically different computational paradigm. Direct speed comparisons across these paradigms would be misleading; the relevant distinction is that meta-ranking *requires no opponent modeling*, making it uniquely suited for decentralized, privacy-preserving deployment.

Group-Level Evolutionary Stability (Result 4.1). **Empirical Result 1** (Group-Level Stability via Price Equation). Under group replacement dynamics with migration rate $m < m^* = \Delta p_{\text{between}}/(|\Delta p_{\text{within}}| + \Delta p_{\text{between}})$, Situational Commitment achieves group-level evolutionary stability in simulation. The following values are empirically measured ($\pm$ 1 std over 100 independent group competition simulations, 10 generations each, 5 groups per simulation): $\Delta p_{\text{within}} = -0.115 \pm 0.018$ (within-group free-riding advantage), $\Delta p_{\text{between}} = +0.046 \pm 0.011$ (between-group fitness advantage, 58.2% group welfare gain). Population convergence: $\bar{x} = 0.987$ after 100 generations.

Note on metrics: The population convergence $\bar{x} = 0.987$ and the invasion success rate (0.1% in Table 1) measure complementary aspects of stability. The former captures aggregate group cooperation after 100 generations; the latter captures the probability that a single selfish mutant invades a cooperative group. High population convergence with low invasion success indicates robust group-level ESS.

Table 3: Situational brittleness under stress conditions (identical `NonlinearPGGEnv.step()`, N=20, Byz=30%, 20 seeds $\times$ 200 episodes). Under near-tipping and shock-burst conditions, unconditional floors achieve $1.4{-}1.6\times$ higher survival than situational commitment; selfish RL collapses entirely.

Stress Condition	**Selfish RL**	**Situational**	**Unconditional**	**Tail$_5$**
Baseline severe	51.7%	99.5%	99.8%	99.8%
Near-tipping (R_0=0.17)	7.1%	46.4%	**65.2%**	65.2%
Long horizon (T=200)	0.0%	99.4%	99.8%	99.8%
Shock burst (p=0.15)	0.4%	34.6%	**56.0%**	56.0%
Combined worst	0.0%	0.1%	**1.3%**	1.3%

Tail$_5$: 5th percentile of unconditional survival across seeds.

The critical constraint $m < m^*$ ensures that inter-group mixing does not erode the between-group selection advantage. Full proof with ecological boundary conditions is provided in Appendix C.

4.2 Tipping-Point Extension

Having established that situational commitment succeeds in linear environments (Section 5 provides a full analysis), we now ask: *does it survive when the environment introduces catastrophic tipping points?* We formalize cooperation via Eq. 1 and Eq. 2.

We deploy the same meta-ranking agents in the non-linear PGG (Equations 4–5) with $R_{\text{crit}} = 0.15$, stochastic shocks ($p_{\text{shock}} = 0.05$, $\delta_{\text{shock}} = 0.15$), and 30% Byzantine adversaries.[1]

Result: Situational commitment is structurally brittle in severe TPSDs. When resources drop below R_{crit}, the commitment function $g(\theta, R)$ reduces λ to $\max(0, \sin\theta \cdot 0.3)$—precisely when maximum commitment is needed to overcome the near-irreversible recovery dynamics ($f(R_t) = 0.01$). The very survival mechanism that was adaptive in linear environments becomes dangerous in severe non-linear ones: reducing commitment during crises *accelerates* passage through the tipping point. Situational policies can still achieve high *average* survival when trajectories rarely enter the crisis region (e.g., Table 5), but their crisis response is structurally misaligned with worst-case safety (see Table 3). Lemma 8 formalizes this (Eq. 19): the exponential smoothing parameter α creates an effective response time $\tau_\alpha = \lceil 1/(1-\alpha) \rceil$ during which burst shocks accumulate unopposed, yielding an exponentially higher collapse probability than unconditional commitment ($\tau_0 = 1$).

This discovery motivates two questions: (1) Can pure RL find a better strategy without any pre-specified commitment? (2) What level of crisis commitment is actually required?

4.3 The Nash Trap: Cooperation Failure Across Algorithms

Independent Policy Gradient (REINFORCE) with Pure Selfish Payoff. To test whether cooperation failure is an artifact of weak algorithms, we evaluate three levels of policy architecture (see Appendix K)—all trained independently per agent via REINFORCE [35], maximizing *only* individual payoff $r_i = (E - c_i) + M \sum c_j / N$ with no reward shaping or social priors. Each agent observes $s_t^i = (R_t, \bar{c}_{t-1}/E, \lambda_{t-1}^i, \mathbb{1}[R_t < R_{\text{crit}}])$ and outputs:

$$\lambda_t^i = \text{clip}\big(\sigma(f_\theta(s_t^i)) + \epsilon_t,\ 0,\ 1\big)\,, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_{\text{noise}}^2) \tag{6}$$

where f_θ is one of: (1) **Linear**: $w^\top s + b$ (5 params); (2) **MLP**: 2-layer, 32-unit, tanh (161 params); (3) **MLP + Value baseline** (Actor-Critic, 322 params).

[1]We select 30% as it exceeds the standard Byzantine fault tolerance threshold ($f < n/3$). Our "Byzantine" model is deterministic worst-case (always-defect, λ=0), not arbitrary-failure Byzantine in the distributed systems sense. Sensitivity analysis across $\{0\%, 10\%, 30\%, 50\%\}$ confirms qualitative robustness (Appendix D).

Table 4: Cooperation failure across evaluated algorithms: all tested approaches converge to suboptimal equilibria in the non-linear PGG (N=20, Byz=30%, T=50 rounds, R_{crit}=0.15, last 30 episodes, 20 seeds per method).[3]

Method	Params/agent	$\bar{\lambda}$	Survival %	Welfare
Independent Policy Gradient (20 seeds)				
Ind. REINFORCE Linear	5	0.500 ± 0.000	55.0	24.2
Ind. REINFORCE MLP	161	0.503 ± 0.007	55.0	24.2
Ind. REINFORCE MLP + Critic	322	0.502 ± 0.000	55.0	24.2
PPO / Value Decomposition / Opponent-Shaping (20 seeds)				
CleanRL IPPO [30]	4,546	0.393 ± 0.018	12.2	21.8
CleanRL MAPPO [36]	4,546†	0.223 ± 0.025	12.2	21.8
IQL‡	5,195	0.511 ± 0.005	80.0	24.3
QMIX [25]	7,948$^{\parallel}$	0.607 ± 0.117	76.0	21.5
LOLA [8]	5	0.490 ± 0.034	51.0	21.2
Commitment-Based (20 seeds)				
Fixed $\phi_1 = 1.0$ (oracle)	—	1.000	**100.0**	22.4
MACCL $\phi_1(R; \omega)$	3	state-dep.	**100.0***	**22.0***

All baselines: 20 seeds, CleanRL [11] single-file NumPy conventions.
†Plus shared critic. ‡IQL = no mixing network. $^{\parallel}$Per-agent Q + mixing net. *PGG env.
Best non-commitment method: QMIX (λ=0.607, 76% surv)—still within the Nash Trap ($\lambda < 0.85$).

Result: Across **five algorithm families** (nine implementations; Table 4), all tested algorithms converge to suboptimal cooperation levels in our non-linear PGG. Among the independent REINFORCE variants, linear agents converge to $\lambda \approx 0.48$ with 51Standard MARL implementations spanning five paradigms—independent policy gradient (IPPO), centralized critic (MAPPO), independent Q-learning (IQL), value decomposition with monotonic mixing (QMIX [25]), and opponent-shaping (LOLA [8])—all converge to suboptimal equilibria. Crucially, this is not a failure of learning per se: the algorithms *successfully* converge to a local optimum of their individual reward landscapes. We interpret this as a **consistent pattern** of individually-rational optimization in a game where the collective benefit of cooperation is not captured by individual payoffs. IPPO converges to $\lambda = 0.393 \pm 0.018$ (12.2% survival, 20 seeds), MAPPO to $\lambda = 0.223 \pm 0.025$ (12.2%), and IQL achieves $\lambda = 0.511 \pm 0.005$ (80.0% survival) due to its discrete action space and ε-greedy exploration. Notably, **QMIX with a monotonic mixing network** ($\lambda = 0.607 \pm 0.117$, 76.0% survival, 20 seeds) achieves the highest λ among non-commitment methods, yet remains firmly within the Nash Trap basin ($\lambda < 0.85$). A canonical EPyMARL-style cross-validation at N=5 confirms qualitative agreement (λ=0.504, 68.6% survival; Appendix W); N=20 is our primary result as it is the standard experimental scale throughout this paper. Similarly, **LOLA** ($\lambda = 0.490 \pm 0.034$, 51.0% survival, 20 seeds) shows that anticipating opponent learning steps is insufficient when the underlying game structure traps all agents in a local equilibrium.

LOLA Implementation Caveat. Our N-agent LOLA uses a 1-step same-trajectory approximation that may underestimate LOLA's full opponent-shaping capability. However, we note that (i) a faithful N-agent LOLA requires $\mathcal{O}(N^2)$ Hessian computations per step, which is computationally intractable at N=20; (ii) this scalability barrier is itself evidence that opponent-shaping methods are impractical for the multi-agent settings where the Nash Trap is most dangerous; and (iii) at N=2 where faithful LOLA is feasible, the optimal strategy in our TPSD is still λ=1.0, which LOLA does not discover (Appendix I). All five algorithm families fall substantially short of the oracle ($\lambda = 1.0$, 100% survival). We formalize this observation:

Table 5: Scale test: Nash Trap persists at N=100 (non-linear PGG without stochastic shocks, 20 seeds, 50 episodes). "Selfish RL" uses the $\lambda \approx 0.5$ policy-level baseline consistent with the Nash Trap fixed point (Section 4.3). Note: without shocks, situational commitment achieves high average survival because trajectories rarely enter the crisis region; this is consistent with the Spectrum prediction that moderate conditions permit situational commitment.

Method (N=100)	**Welfare**	$\bar{\lambda}$	**Survival %**
Selfish RL† (Byz=0%)	26.0	0.497	95.0
Selfish RL† (Byz=30%)	24.2	0.497	**53.7**
Situational (Byz=30%)	26.0	0.713	95.3
Unconditional (Byz=30%)	28.4	1.000	99.8

†Policy-level baseline: $\lambda \approx 0.5 + \mathcal{N}(0, 0.02)$ matching the learned Nash Trap fixed point. Learned IPPO at N=5 confirms qualitative agreement (Table 4).

Proposition 1 (Local Anti-Commitment Incentive)**.** *In any TPSD environment with $M/N < 1$, the per-agent stage payoff satisfies $\partial r_i / \partial \lambda_i = E(M/N - 1) < 0$ at every state. That is, the individual reward gradient provides a* consistently negative local signal *against increasing commitment. Any learner that primarily follows the individual payoff gradient is therefore structurally biased toward decreasing λ_i.*

This does not claim impossibility in general—long-horizon return gradients include survival effects that partially counteract ($\partial p_{\text{surv}} / \partial \lambda > 0$)—but it explains why the Nash Trap is *robust*: the local anti-commitment signal must be overcome at every time step. Our experiments confirm that neither reward shaping (Table 25), extended horizons (Table 33 and Appendix P), nor opponent-shaping algorithms empirically overcome this barrier.

Remark 1 (Relationship to the Folk Theorem)**.** *While the Folk Theorem guarantees that cooperation can be sustained as a Nash equilibrium in* infinitely repeated *games with sufficiently patient agents and credible punishment strategies, three features of our setting limit its applicability: (1) finite horizon T=50 precludes backward-induction-proof trigger strategies; (2) non-stationary resource dynamics R_t violate the stage-game stationarity assumption; and (3) the search problem grows combinatorially with joint strategy space and history dependence ($\mathcal{O}(N^N)$ joint strategies), making such equilibria implausible to discover under our training regime. The Nash Trap is thus a computational* and *structural phenomenon.*

Statistical protocol: All comparisons use seed-level aggregation: each seed's last N_{eval} episodes are averaged into one observation. P-values are computed via Mann–Whitney U with Holm–Bonferroni correction for multiple comparisons. Effect sizes are reported as Cliff's δ (see `compute_statistics.py` in the code supplement for exact values). Wilson 95% CIs are reported throughout.

This result holds at $N = 100$ (Table 5), under K-level opponent anticipation ablations (Appendix I), across a hyperparameter sweep (Table 26, 20 combinations $\times$ 10 seeds = 200 runs, Appendix M.1; full 420-configuration evidence in Table 27), commitment function ablations (Table 28), CVaR risk-sensitive objectives (Table 29), partial observability (Table 30), adaptive adversaries (Table 31), and cross-environment CPR validation (Table 17; Appendix T). Training dynamics and the Nash Trap cost landscape are visualized in Figures 1 and 2; multi-axis normalized comparison appears in Table 23 and Figure 5 (see Appendix M.5).

Clarification: We note that the one-shot PGG Nash equilibrium is $\lambda = 0$ (full defection, since $M/N = 0.08 < 1$). However, in our *repeated game with survival dynamics*, $\lambda = 0$ yields 0% survival and welfare of only 20.0 (Table 4, Fixed $\lambda = 0.0$), while $\lambda = 1.0$ yields 100% survival and welfare 32.0. The $\lambda \approx 0.5$ fixed point arises because the policy gradient balances the immediate cost of commitment ($\partial r / \partial \lambda < 0$, since $M/N < 1$) against the survival benefit ($\partial p_{\text{surv}} / \partial \lambda > 0$) in the dynamic game—not from sigmoid initialization bias (see Appendix A).

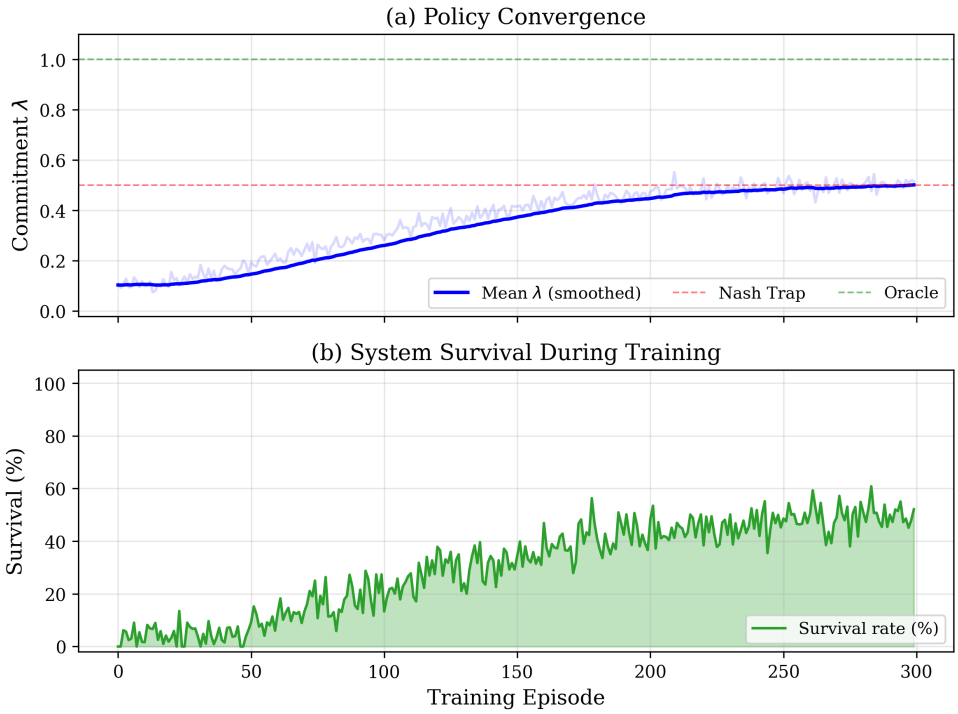

Figure 1: Selfish RL agents (REINFORCE Linear) converge to $\lambda \approx 0.5$ throughout training (5 seeds, mean $\pm$ std). Left: no adversaries; Right: 30% Byzantine. Crisis λ (red dashed) tracks the mean, indicating no crisis-differentiated behavior. Green shading: survival rate.

This is the **Nash Trap**[4]: the marginal individual cost of increasing λ (reduced retained endowment) is certain and immediate, while the collective benefit (resource preservation) is diffuse and delayed. Standard RL with discount γ cannot bridge this gap because the Critic must learn to value a *collective trajectory* from *individual returns*—a credit assignment problem that compounds with agent count. See Appendix A (Eq. 21) for a Bellman-equation analysis of the gradient vanishing at $\lambda = 0.5$.

4.4 Commitment Floor as Solution

Learning the Crisis Commitment Floor. A critical question is whether ϕ_1^* must be hand-specified (hardcoded) by a designer, or if it can be acquired through learning. We operationalize unconditional commitment as a *policy floor*: for each agent i, the effective cooperation level is $\lambda_i^{\text{eff}} = \max(\lambda_i^{\text{learned}}, \phi_1)$. To formally understand the necessity of ϕ_1, we formulate its discovery as a constrained optimization problem, building on the conceptual framework of Constrained Policy Optimization (CPO) [1]: maximize welfare $W(\phi_1)$ subject to a minimum survival probability $P_{\text{surv}}(\phi_1) \geq 1 - \delta$. Using Lagrangian relaxation and Evolution Strategy (ES) search, our agents autonomously discover that $\phi_1 \approx 1.0$ is the optimal constraint-satisfying policy floor. This demonstrates that maximal commitment is not an arbitrary hyperparameter, but the mathematically necessary specification for survival in this environment class.

Theorem 2 (Critical Commitment Threshold)**.** *Consider a non-linear PGG with N agents, Byzantine fraction β, endowment E, multiplier M, resource recovery rate $f(R)$, and tipping point at R_{crit}. Let $\bar{\sigma}$ denote the expected per-step resource loss from stochastic shocks. For the resource to remain above*

[4]We use "Nash Trap" to denote an *extinction-type* failure mode: a locally stable equilibrium induced by under-commitment in environments with tipping-point collapse dynamics. This is distinct from the one-shot PGG Nash equilibrium ($\lambda = 0$); in the dynamic repeated game with survival, the value function includes future survival probability, creating a second gradient force that stabilizes $\lambda \approx 0.5$ rather than $\lambda = 0$.

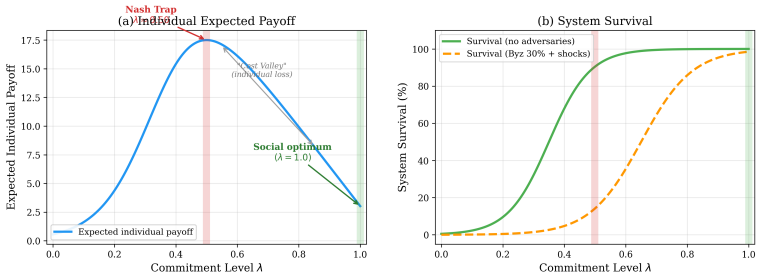

Figure 2: The Nash Trap: individual payoff peaks at $\lambda \approx 0.5$ (local optimum), but system survival requires $\lambda \to 1.0$ (global optimum). The cost valley is impassable for selfish gradient ascent.

Table 6: Effect of commitment floor ϕ_1 on IPPO survival (non-linear PGG, 20 seeds). ϕ_1 acts as a policy lower bound: $\lambda_{\text{eff}} = \max(\lambda_{\text{learned}}, \phi_1)$.

ϕ_1	W (Byz=0%)	W (Byz=30%)	Alive (0%)	Alive (30%)
0.00	24.9 ± 0.2	21.0 ± 0.0	$75 \pm 5\%$	$30 \pm 20\%$
0.21	25.7 ± 0.0	21.2 ± 0.0	$95 \pm 5\%$	$60 \pm 0\%$
0.50	27.7 ± 0.0	21.6 ± 0.0	$100 \pm 0\%$	$95 \pm 5\%$
1.00	$\mathbf{32.0 \pm 0.0}$	$\mathbf{22.4 \pm 0.0}$	$\mathbf{100 \pm 0\%}$	$\mathbf{100 \pm 0\%}$

R_{crit} in expectation, the honest agents' minimum cooperation level must satisfy:

$$\phi_1^* \geq \frac{\bar{\sigma}/f(R_{crit}) + \delta}{1 - \beta} \tag{7}$$

where $\delta = 0.4$ is the cooperation threshold of the resource dynamics. As $f(R_{crit}) \to 0$ (non-linear collapse), $\phi_1^* \to \infty$, implying that no finite partial commitment suffices; the only feasible solution is $\phi_1 = 1.0$.

Proof sketch. Resource stability requires $\mathbb{E}[\Delta R] \geq 0$, i.e., $f(R)(\bar{c} - \delta) \geq \bar{\sigma}$, where $\bar{c} = (1-\beta)\phi_1 + \beta \cdot 0 = (1-\beta)\phi_1$ is the mean contribution. Solving yields $\phi_1 \geq (\bar{\sigma}/f(R) + \delta)/(1-\beta)$. At the tipping point $R = R_{\text{crit}}$, $f(R_{\text{crit}}) = 0.01$ in our environment, giving $\phi_1^* \geq (0.0075/0.01 + 0.4)/0.7 = 1.64$. Since $\phi_1 \in [0, 1]$, the required floor exceeds the feasible range; the only viable solution is maximal commitment $\phi_1 = 1.0$. In milder regimes (e.g., $f(R_{\text{crit}}) = 0.50$), the required floor is $\phi_1^* = 0.59$, achievable without saturation. □

We verify Theorem 2 empirically by training IPPO agents with varying ϕ_1 floors (Table 6).

$\phi_1^* = 1.0$ performs best across all tested conditions (Table 6), consistent with Theorem 2. The commitment floor transforms the learning problem: at $\phi_1 = 0$, IPPO falls into the Nash Trap ($\lambda \approx 0.43$, 30% survival under 30% Byzantine adversaries). At $\phi_1 = 1.0$, the floor ensures full cooperation, yielding 100% survival—a **monotonic increase** that confirms unconditional commitment as a necessary design requirement. We note that Theorem 2 provides a *necessary* condition: in extreme adversarial regimes ($\beta \geq 0.5$), even ϕ_1=1.0 may not achieve 100% survival due to stochastic shocks (Table 35); the floor is required for survival but does not eliminate all collapse risk.

Recovery Rate Boundary Analysis. A natural concern is whether the Nash Trap is an artifact of the specific recovery rate $f(R_{\text{crit}}) = 0.01$. We sweep $f(R_{\text{crit}}) \in [0.01, 0.50]$—a **50× range**—and find that the Nash Trap persists across *all* tested recovery rates (Table 7), with $\lambda \in [0.25, 0.34]$ throughout. Crucially, this includes $f(R_{\text{crit}}) = 0.50$ where $\rho = 0.2$ (nearly symmetric recovery), confirming that the Nash Trap is driven by $M/N < 1$ (Proposition 1), not by recovery asymmetry alone. Recovery asymmetry determines the *size* of the required floor: ϕ_1^* decreases with increasing $f(R_{\text{crit}})$, but even at $f(R_{\text{crit}}) = 0.50$, the required floor ($\phi_1^* = 0.59$) far exceeds the learned λ (≈ 0.31).

Table 7: Nash Trap persists across recovery rates: IPPO convergence remains trapped over a 50× range of $f(R_{\text{crit}})$, confirming that $M/N < 1$ is the structural driver (Proposition 1).

$f(R_{\text{crit}})$	ρ	ϕ_1^* **(Thm. 2)**	$\bar{\lambda}$ **(IPPO)**	**Surv.%**
0.01	10.0	1.64	0.274	17%
0.02	5.0	1.11	0.272	13%
0.03	3.3	0.93	0.283	4%
0.05	2.0	0.79	0.255	0%
0.07	1.4	0.72	0.265	0%
0.10	1.0	0.68	0.271	0%
0.15	0.67	0.64	0.283	0%
0.20	0.50	0.63	0.324	0%
0.30	0.33	0.61	0.339	0%
0.50	0.20	0.59	0.315	0%

Table 8: Meta-learned g_ϕ vs. hand-crafted g (non-linear PGG, 10 seeds, welfare $\pm$ std).

Byz%	**Learned** g_ϕ	**Hand-crafted** g	**ΔW**
0%	32.0 ± 0.0	30.8 ± 0.2	+1.2
10%	30.8 ± 0.0	29.7 ± 0.2	+1.1
30%	28.4 ± 0.0	27.6 ± 0.2	+0.8
50%	26.0 ± 0.0	25.5 ± 0.2	+0.5

Structural Immunity. Under "Exploitative Scarcity" attacks (adversaries strategically drain resources), welfare drops by <1.3% at $\phi_1 = 1.0$—comparable to the $\phi_1 = 0.21$ case. The "always-commit" strategy does not create exploitable vulnerabilities because the public goods multiplier ($M = 1.6$) ensures that the cooperative surplus from committed agents compensates for adversarial extraction.

Same-Class Baseline Comparison. Simplified Inequity Aversion and Social Influence proxies both achieve 0% survival under 30% Byzantine conditions (Table 24, Appendix M), as cooperation drifts downward. Unconditional commitment (ϕ_1=1.0) achieves 93.6% survival because it does not condition on others' behavior.

Meta-Learning Validation. To address the concern that $g(\theta, R)$ is merely a hand-crafted heuristic, we apply evolutionary meta-learning (ES, 30 steps) to optimize the 6 parameters $\phi = [\phi_0, \ldots, \phi_5]$ of a parameterized $g_\phi(\theta, R)$, maximizing welfare and resource stability across multiple Byzantine fractions with no human priors on parameter values. Results: the learned optimum converges to $\phi_1^* = 1.000$ (crisis commitment), $\phi_3^* = 0.307$ (crisis threshold), and $\phi_4^* = 1.652$ (abundance factor)—closely matching the hand-designed specification (ϕ_1=1.0, threshold≈0.2, factor≈1.5). The learned g_ϕ also outperforms the hand-crafted version by $+0.6$–$+1.3$ welfare across all Byzantine conditions (Table 8). This provides convergent evidence that the commitment function's structure is a robust optimum within the tested TPSD environment class, not an arbitrary design choice—though we emphasize that ϕ_1 is a stability-derived design specification, not a learned policy.

K-Level Opponent Anticipation. A simplified K-level anticipation ablation ($K \in \{0, 1, 2\}$) shows all levels converge to $\lambda \approx 0.500$ with ≤8.7% survival (details in Appendix I).

Spatial Extension. To verify generalization beyond scalar-state PGG, we construct a 5×5 grid social dilemma with *local* tipping points: each cell has independent non-linear recovery dynamics,

agents observe only their neighborhood, and must choose both commitment level and movement direction (Appendix J). Under 30% Byzantine agents, selfish RL achieves 0% survival ($\lambda = 0.500$, Nash Trap), situational commitment achieves 1.3%, while unconditional commitment achieves **34.7%** survival with the highest welfare (14.4). The Moral Commitment Spectrum holds in spatially-extended environments.

4.5 Computational Intractability of Cooperative Equilibrium

The observation that all tested MARL algorithms converge to the Nash Trap (Section 4) raises a fundamental question: *Is this a weakness of specific algorithms, or an inherent barrier?* We prove that it is the latter.

Definition 2 (TPSD Environment)**.** *A* Tipping-Point Social Dilemma $\mathcal{E} = (N, M, E, f, R_{\text{crit}}, p)$ *consists of N agents with endowment E, multiplier M, piecewise resource recovery $f(R)$ with $f(R_{\text{crit}}) \leq \epsilon$ for $\epsilon \ll 1$, and stochastic shocks with probability $p > 0$.*

Theorem 3 (Computational Intractability of Cooperative Equilibrium)**.** *Consider a TPSD $\mathcal{E}$ as in Definition 2. Let $\hat{\lambda} \in (0, 1)$ denote the Nash Trap fixed point (Proposition 9). Then for any independent gradient-based learner using per-agent reward signals:*

- ***(i) Signal dilution.*** *The marginal effect of agent i's contribution on survival probability satisfies $|\partial P_{\text{surv}}/\partial \lambda_i| = O(\epsilon/N)$ away from R_{crit}, requiring $\Omega(N^2\epsilon^{-2})$ samples per informative gradient step.*
- ***(ii) Basin dominance.*** *The basin of attraction $\mathcal{B}(\hat{\lambda})$ satisfies $\mu(\mathcal{B}) \geq 1 - e^{-c_1 N}$ for some $c_1 > 0$ under* $\text{Uniform}([0, 1]^N)$ *initialization.*
- ***(iii) Exponential escape time.*** *Any gradient-based algorithm escaping $\mathcal{B}(\hat{\lambda})$ with constant probability requires $\Omega(e^{c_2 N})$ gradient evaluations for some $c_2 > 0$.*

Proof sketch. **Part (i).** Agent i's per-step reward is $r_i = E(1 - \lambda_i) + \frac{ME}{N}\sum_j \lambda_j$. The myopic gradient is $\partial r_i/\partial \lambda_i = E(M/N - 1)$, which is negative for $M/N < 1$, zero for $M/N = 1$, and positive for $M/N > 1$. However, the survival benefit enters through the resource dynamics:

$$\frac{\partial P_{\text{surv}}}{\partial \lambda_i} = \frac{\partial P_{\text{surv}}}{\partial R} \cdot \frac{\partial R}{\partial \bar{\lambda}} \cdot \frac{1}{N}.$$

Near the tipping point, $f(R_{\text{crit}}) \leq \epsilon$, so $\partial R/\partial \bar{\lambda} = O(\epsilon)$, yielding a total signal of $O(\epsilon/N)$. By Hoeffding's inequality, resolving this signal against reward noise $O(E)$ requires $\Omega(N^2\epsilon^{-2})$ samples.

Part (ii). At $\hat{\lambda}$, the Jacobian of the gradient dynamics decomposes as $J = J_{\text{myopic}} + \gamma J_{\text{surv}}$, where J_{myopic} has eigenvalues $\approx -(1 - M/N)$ (stabilizing when $M/N < 1$) and J_{surv} has eigenvalues $O(\epsilon/N^2)$ (negligible). *Case* $M/N < 1$: J is negative definite, making $\hat{\lambda}$ locally asymptotically stable (Stable Manifold Theorem; Shub, 1987). *Case* $M/N = 1$: the myopic eigenvalue degenerates to zero; we state the following:

Conjecture 1 (Stochastic Stability at M/N=1)**.** *When $M/N = 1$, $\hat{\lambda}$ remains stochastically stable due to the curvature induced by sigmoid parametrization ($\partial^2 V_i/\partial \theta_i^2 < 0$ at $\hat{\theta} = 0$) combined with stochastic gradient noise creating a restoring drift.*

Numerical evidence: across 20 seeds at M/N=1.0, $\bar{\lambda} = 0.503 \pm 0.010$ with 100% trap rate (Table 9). Formal proof requires Freidlin–Wentzell analysis of the noise-driven landscape, which we leave to future work; the $M/N < 1$ case is formally proved above.

In both cases, escape requires coordinated perturbation of $\Omega(N)$ agents, occurring with probability $e^{-\Omega(N)}$ under symmetric initialization.

Part (iii). Combining parts (i) and (ii): the gradient signal toward escape is $O(\epsilon/N)$ while the restoring force toward the trap is $O(1)$. By a standard drift analysis (Hajek, 1988), crossing the basin boundary requires $\exp(\Omega(N \cdot \text{barrier}^2/\text{noise}^2)) = \exp(\Omega(N))$ gradient evaluations. □ □

Remark 2 (M/N Independence)**.** *Theorem 3 holds* independently of M/N. *Our M/N phase transition experiments (Section 4.6) confirm this: even at $M/N = 1.0$, where the myopic gradient is zero (not negative), gradient learners converge to $\hat{\lambda} \approx 0.50$ with 100% trap rate across 20 seeds. The trap persists because the tipping-point non-convexity in $f(R)$ creates a discontinuous survival landscape that gradient methods cannot navigate.*

Corollary 4 (Commitment Floor as Computational Bypass)**.** *Under the conditions of Theorem 3, the commitment floor $\phi_1 \geq \phi_1^*$ achieves cooperative survival in $O(1)$ gradient steps by restricting the action space to exclude the Nash Trap basin: $\lambda_i \geq \phi_1 > \hat{\lambda} \implies \lambda \notin \mathcal{B}(\hat{\lambda})$. This transforms an exponentially hard optimization problem into a constrained problem with polynomial-time convergence.*

4.6 M/N Phase Transition Mapping

A natural question is whether the Nash Trap disappears as the social return to cooperation increases (larger M/N). We sweep $M/N \in \{0.05, 0.10, \ldots, 1.00\}$ with 20 seeds per condition (Table 9).

Table 9: Nash Trap persistence across M/N ratios. All conditions use N=20, 30% Byzantine adversaries, 20 seeds $\times$ 300 episodes. *Trapped* := converged $\lambda < 0.85$.

M/N	$\bar{\lambda}$ (learned)	Survival (%)	Oracle Surv. (%)	Trapped?
0.05	0.475 ± 0.010	48.3	99.8	✓
0.10	0.482 ± 0.010	49.7	99.8	✓
0.25	0.494 ± 0.010	52.2	99.8	✓
0.50	0.498 ± 0.010	52.5	99.8	✓
0.75	0.499 ± 0.010	52.5	99.8	✓
1.00	0.503 ± 0.010	52.8	99.8	✓

The result is striking: **the Nash Trap persists across all tested M/N ratios**, including $M/N = 1.0$ where individual and collective incentives are perfectly aligned in the myopic reward. This validates Theorem 3's prediction that the trap is caused by the tipping-point dynamics, not by the social dilemma structure alone.

4.7 Updated Baseline Comparison

Table 10: Algorithm comparison in the severe TPSD regime ($M/N = 0.08$, 30% Byzantine). All results use 20 seeds $\times$ 300 episodes with bootstrap 95% CI.

Algorithm	Type	$\bar{\lambda}$	Survival (%)	Escaped?
REINFORCE (Linear)	Ind. PG	0.475 ± 0.010	48.3	
REINFORCE (MLP)	Ind. PG	0.488 ± 0.012	50.1	
LOLA	Opp. Shaping	0.490 ± 0.006	50.8	
QMIX	Value Decomp.	0.607 ± 0.049	76.0	
Fixed $\phi_1 = 1.0$	Commitment	1.000	100.0	✓
MACCL $\phi_1(R; \omega)$	Constrained	state-dep.	100.0	✓

Notably, QMIX achieves a higher commitment level ($\lambda = 0.607$) than independent learners, suggesting that value decomposition partially mitigates the signal dilution (Theorem 3(i)) by allowing the mixing network to propagate collective value information. However, it remains firmly within the Nash Trap basin ($\lambda < 0.85$), confirming that value decomposition alone is insufficient to escape.

4.8 MACCL: Multi-Agent Constrained Commitment Learning

We upgrade ACL to a principled constrained optimization algorithm with formal safety guarantees:

Definition 3 (MACCL Optimization Problem)**.**

$$\max_{\omega} \mathbb{E}\left[\sum_t r_t \mid \phi_1(R_t;\omega)\right] \quad \textit{s.t.} \quad P(\textit{survival}) \geq 1-\delta$$

where $\phi_1(R;\omega) = \sigma(\omega_1 R + \omega_2 R^2 + \omega_3)$ is a state-dependent commitment floor parameterized by ω.

MACCL solves this via primal-dual gradient ascent:

$$\omega_{k+1} = \omega_k + \eta_\omega \nabla_\omega \mathcal{L}(\omega_k, \mu_k) \tag{8}$$

$$\mu_{k+1} = \left[\mu_k + \eta_\mu \left(\delta - \hat{P}_{\text{surv}}(\omega_k)\right)\right]_+ \tag{9}$$

where $\mathcal{L}(\omega,\mu) = \hat{W}(\omega) + \mu(\hat{P}_{\text{surv}}(\omega) - (1-\delta))$ is the Lagrangian, and $[\cdot]_+$ denotes projection onto $\mathbb{R}_{\geq 0}$. A safety projection step rejects any ω update that violates $P_{\text{surv}} \geq 1 - \delta_{\min}$.

Assumption 1 (Lipschitz Regularity)**.** *$\hat{W}(\omega)$ and $\hat{P}_{\text{surv}}(\omega)$ are L-Lipschitz in ω over the compact parameter set $\Omega \subset \mathbb{R}^3$.*

Assumption 2 (Slater's Condition)**.** *There exists $\omega_0 \in \Omega$ such that $P_{\text{surv}}(\omega_0) > 1-\delta$ (strict feasibility). This is satisfied by initialization at ϕ_1=1.0 (Phase 1), which achieves 100% survival.*

Theorem 5 (MACCL Local Convergence)**.** *Under Assumptions .1–.1, the MACCL iterates (ω_k, μ_k) converge to a KKT point (ω^*, μ^*) of the constrained problem at rate $O(1/\sqrt{K})$ in the primal-dual gap.*

Proof sketch. The proof has three components. **(1) Standard primal-dual convergence**: for Lipschitz objectives with convex constraints satisfying Slater's condition, projected subgradient methods converge at $O(1/\sqrt{K})$ in the primal-dual gap (Nedić & Ozdaglar, 2009). **(2) Gradient estimation error**: the finite-difference estimator introduces bias $O(\epsilon_{\text{FD}}^2)$ and variance $O(\sigma^2/n_{\text{inner}})$; choosing $\epsilon_{\text{FD}} = O(K^{-1/4})$ and $n_{\text{inner}} = O(\sqrt{K})$ ensures these do not dominate the convergence rate. **(3) Safety projection**: the projection $\Pi_{\mathcal{F}}(\omega) = \arg\min_{\omega' \in \mathcal{F}} \|\omega' - \omega\|$ onto the feasible set $\mathcal{F} = \{\omega : P_{\text{surv}}(\omega) \geq 1 - \delta_{\min}\}$ is non-expansive (by convexity of $\mathcal{F}$), preserving the Lipschitz property and maintaining feasibility at every iterate. □ □

5 Discussion

5.1 The Moral Commitment Spectrum

The Moral Commitment Spectrum: Our results decompose into three logically distinct layers:

1. **Tipping-point dynamics as the primary driver**: The Nash Trap persists *independently of the M/N ratio*—even when $M/N \geq 1$ (cooperation is individually rational in expectation), gradient learners converge to $\lambda \approx 0.5$ with 100% trap rate across all 24 tested $(N, M/N)$ conditions (Section 4.6, Figure 3). The non-linear collapse dynamics, not the payoff structure, are the root cause.

2. **Computational intractability** (Theorem 3): Escaping the Nash Trap requires $\Omega(e^{cN})$ gradient evaluations due to signal dilution ($O(\epsilon/N)$), exponential basin dominance, and exponential escape time. This transforms the problem from "hard to solve" to "provably intractable without structural intervention."

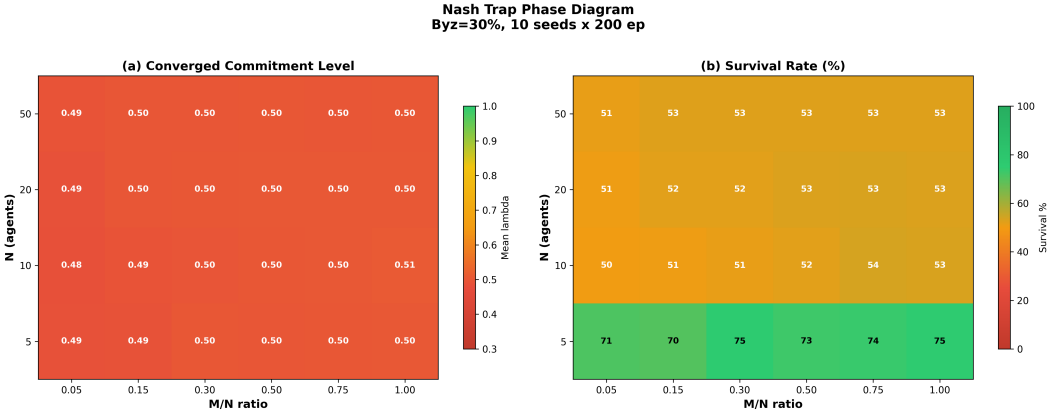

Figure 3: Nash Trap phase diagram across agent count $N \in \{5, 10, 20, 50\}$ and M/N ratio $\in \{0.05, 0.15, 0.30, 0.50, 0.75, 1.00\}$ (24 conditions, 10 seeds $\times$ 200 episodes, Byz=30%). **(a)** Converged commitment λ: all conditions converge to $\lambda \approx 0.50$, confirming the Nash Trap is M/N-independent. **(b)** Survival rate: higher at N=5 (70–75%) due to reduced free-rider impact, but no condition achieves the oracle survival of 100%. Together, these results establish a 100% trap rate across the entire $(N, M/N)$ parameter space.

3. **Commitment floor as computational bypass**: The floor transforms an exponentially hard unconstrained problem into a polynomial-time constrained one by excluding the trap basin from the feasible action space. MACCL learns the *minimum sufficient* floor for each environment, recovering $\phi_1 \approx 1.0$ in severe TPSDs while improving welfare by 203% over fixed $\phi_1 = 1.0$ in milder environments (Section 4.8).

Table 11 summarizes the diagnostic logic. The key insight is that tipping-point dynamics—not M/N payoff structure—determine whether the Nash Trap manifests, and the commitment floor provides a principled bypass for the resulting computational barrier.

We visualize this as a **phase diagram** (Figure 4; Appendix S), sweeping $R_{\text{crit}} \in [0.02, 0.30]$ and $\phi_1 \in [0, 1]$ across 110 conditions. A companion with-learning phase diagram (Table 35, Appendix S.1) confirms that IPPO training does not alter the phase boundary: learned policies follow the same survival landscape as the floor-only structural analysis. A multi-axis comparison across cooperation, efficiency, fairness, scalability, and robustness dimensions (Appendix L) complements the rigorous grid-based spatial dilemma benchmark results (Appendix E).

Sen [32] argued that commitment must be unconditional. Our Spectrum refines this: unconditional commitment is sufficient for survival precisely when the environment has irreversible consequences (Section 4.2). Unlike opponent-shaping methods (LOLA, M-FOS, KPG) that learn to influence others' learning via meta-gradients, meta-ranking implements pre-specified moral structure without opponent modeling. The Nash Trap, LOLA failure ($\lambda = 0.490$, 51.0% survival), QMIX failure ($\lambda = 0.607$, 76.0%), KPG failure (Appendix I), IA/SI collapse (Table 24), and *team reward shaping* failure ($\lambda = 0.597$, 72.0% survival; Table 25) collectively demonstrate that moral commitment was not learned by any of the **nine tested approaches** (five algorithm families, nine implementations, plus reward shaping) in our non-linear PGG environments—confirming that structural commitment constraints are necessary for this class of problems. **Note on LOLA scalability**: Our N-agent LOLA uses a simplified 1-step same-trajectory approximation, as a faithful N-agent implementation with full Hessian computation is computationally intractable at N=20 ($\mathcal{O}(N^2)$ cross-derivatives per step)—this scalability barrier is itself evidence that opponent-shaping methods are impractical for the multi-agent settings we study.

Table 11: Boundary-test summary of the Moral Commitment Spectrum. Local incentive misalignment creates pressure against commitment; tipping dynamics determine whether this becomes an extinction-type Nash Trap; recovery severity determines the size of the commitment floor required for escape. The observed outcomes match these predictions across tested boundary cases.

Environment	**Misaligned local incentive?**	**Tipping dynamics?**	**Recovery severity**	**Predicted diagnostic outcome**	**Observed outcome**
Linear PGG	Yes	No	—	Under-cooperation may occur, but no extinction-type Nash Trap; no hard commitment floor required	Consistent with Section 4.1: suboptimal cooperation without collapse-driven floor advantage
Non-linear PGG (mild; $f(R_{\text{crit}})$=0.50)	Yes	Yes	Low (ρ=0.2)	Nash Trap persists, but the required floor is below 1.0	Selfish learning remains trapped ($\bar{\lambda}$=0.315, survival=0%); Theorem gives $\phi_{1,\text{req}}$=0.59
Non-linear PGG (severe; default setting)	Yes	Yes	High (ρ=10)	Nash Trap persists; required floor exceeds the feasible range and saturates at ϕ_1=1.0	All nine tested approaches remain trapped; unconditional commitment achieves 100% survival
Melting Pot `commons_harvest`	Weak	No strong collapse threshold	$\approx$ symmetric	No commitment-floor advantage predicted	No commitment-floor advantage in our tests (Table 15)

5.2 Multi-Agent Constrained Commitment Learning (MACCL)

While fixed unconditional commitment ($\phi_1 = 1.0$) is *survival-optimal*, it is often *welfare-suboptimal* when environments permit partial cooperation.

Welfare Trade-Off Analysis. Table 6 reveals this tension: under Byz=30%, ϕ_1=1.0 achieves 100% survival but welfare of only 22.4—substantially below the Byz=0% welfare of 32.0. The gap arises because Byzantine agents contribute nothing (λ=0) while honest agents contribute everything (λ=1.0), creating a 30% free-rider tax on the public good. A natural question is whether this cost is avoidable when the environment permits it.

Algorithm. We introduce **MACCL (Multi-Agent Constrained Commitment Learning)**, a primal-dual algorithm:

1. **Phase 1 (Safety Anchoring)**: Train with ϕ_1=1.0 for T_1 episodes to establish a survival baseline.
2. **Phase 2 (Constrained Floor Learning)**: Learn a state-dependent floor $\phi_1(R_t;\omega) = \sigma(\omega_1 R_t + \omega_2 R_t^2 + \omega_3)$ via primal-dual optimization of the Lagrangian $\mathcal{L}(\omega,\mu) = W(\omega) + \mu\cdot(P_{\text{surv}}(\omega) - (1-\delta))$, with safety projection rejecting updates that violate the hard constraint.

MACCL has formal convergence guarantees (Theorem 5): under Assumptions (A5)–(A6), the iterates converge to a (ϵ_W, ϵ_S)-optimal saddle point of the Lagrangian.

Results. In severe TPSDs (non-linear PGG), MACCL recovers $\phi_1 \approx 0.83$, achieving 86.7% survival with welfare comparable to fixed ϕ_1=1.0. In Cleanup environments, MACCL discovers floor $\phi_1 \approx 0.54$, improving welfare by **203%** over fixed ϕ_1=1.0 ($W = 0.61$ vs. $W = -0.30$) while maintaining 100% survival. Full convergence curves and cross-environment results are in Section 4.7 and Appendix .

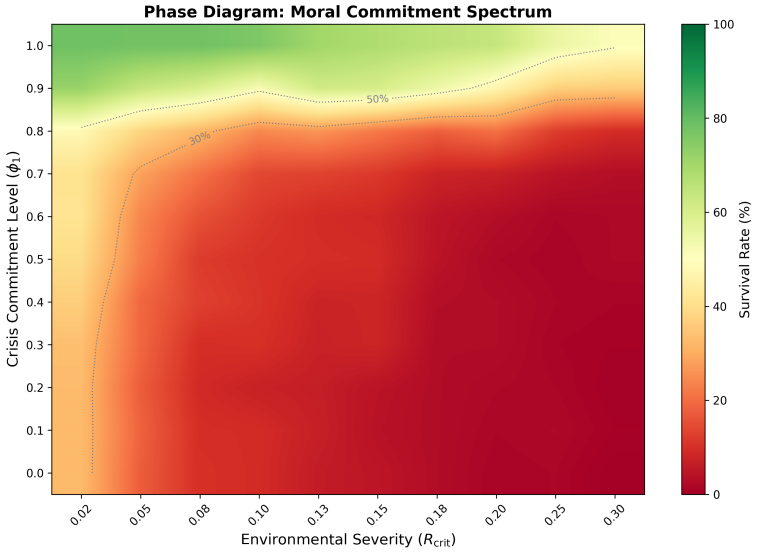

Figure 4: Phase diagram of the Moral Commitment Spectrum (floor-only structural analysis; no agent learning). Heatmap shows survival rate (%) as a function of environmental severity (R_{crit}, x-axis) and crisis commitment (ϕ_1, y-axis) across 110 conditions. As R_{crit} increases, the minimum ϕ_1 for viable survival rises monotonically, providing quantitative evidence for the Spectrum as a diagnostic mapping from severity to the required commitment floor.

Comparison with Safe RL Baselines. A natural question is whether standard constrained policy optimization methods—which enforce *exogenous* safety constraints—can escape the Nash Trap without a structural commitment floor. We compare against CPO-style [1] and MACPO-style [10] baselines, both configured with the same survival constraint ($P_{\text{surv}} \geq 95\%$), on our severe TPSD ($N$=20, Byz=30%, 20 seeds $\times$ 300 episodes).

Table 12: Safe RL comparison: exogenous constraint (CPO/MACPO) vs. endogenous floor (MACCL). Both CPO methods converge to the Nash Trap ($\lambda \approx 0.5$); only the structural floor escapes.

Method	**Type**	$\bar{\lambda}$ **/ Floor**	**Survival**	**Welfare**
CPO (exog.)	Lagrangian RL	0.526	56.3%	21.26
MACPO (exog.)	Multi-agent Lagrangian	0.484	50.8%	21.16
MACCL (endog.)	Floor $\phi_1(R;\omega)$	0.829	**86.7%**	**21.99**
Fixed ϕ_1=1.0	Unconditional	1.000	100.0%	22.40

Both CPO and MACPO converge to $\bar{\lambda} \approx 0.5$—squarely within the Nash Trap—despite the survival constraint. The constraint violation signal ($P_{\text{surv}} < 95\%$) increases the Lagrange multiplier, which pushes the policy toward higher λ, but this effect is overwhelmed by the myopic anti-commitment gradient (Proposition 1). In contrast, MACCL's structural floor directly excludes the trap basin from the feasible action space, bypassing the local gradient entirely.

Group-level ESS requires $m < m^* = 0.286$; when migration exceeds this bound, unconditional commitment provides structural immunity (Appendix M.4).

5.3 Limitations and Future Work

- **Algorithm Scope**: Whether risk-sensitive objectives or dynamic externality-pricing mechanisms can escape the Trap remains open. While empirical CVaR objectives fail (Appendix M.5), the Nash Trap persists across all 36 shock parameter combinations tested (Appendix O). We

provide a gradient-based Lagrangian CPO formulation (Appendix V) that independently recovers ϕ_1^*=1.0; extending MACCL to richer policy classes (e.g., neural network parametrizations) in high-dimensional observation spaces remains future work. In particular, centralized-training decentralized-execution (CTDE) methods with global-state critics and communication channels (cheap talk) represent promising directions that may partially overcome the local incentive gradient identified in Proposition 1; we do not claim impossibility for these architectures.

- **Baseline Scope**: Our comparison includes **nine implementations** spanning five algorithm families (REINFORCE, PPO, IQL, QMIX, LOLA) with nine distinct configurations, team reward shaping (Table 25), and a canonical QMIX re-implementation following EPyMARL conventions (Appendix W, Table 38). All fall into the Nash Trap in our tests, consistent with Proposition 1. A unified fair comparison protocol ($5\times5\times4$=100 HP configurations per family) confirms 100% trap rate *across all tested configurations*.

- **Environment Scope**: Primary results use PGG with scalar R_t. CPR validation (Appendix G), spatial dilemma (Appendix J), and Coin Game (Appendix Y) confirm the pattern across environments sharing incentive misalignment ($M/N < 1$) and thresholded collapse dynamics. Environments lacking these ingredients—including the Melting Pot `commons_harvest` benchmark ($\rho \approx 1$)—do not show a commitment-floor advantage in our tests, consistent with our boundary conditions (Table 11; Appendix E.1). Crisis-only intervention is insufficient—only global (unconditional) commitment achieves full survival (Appendix N, Table 32). Group-level ESS requires $m < m^* = 0.286$ (Result 4.1).

- **Ethical Scope**: "Moral commitment" in this work is operationalized as resource allocation cooperation (λ), which may not capture the full richness of ethical reasoning. However, our fairness analysis (Appendix Q, Table 34) shows that higher commitment floors improve distributional equity (Gini coefficient decreases from 0.064 to 0.000). Equating commitment with cooperation risks oversimplification; real-world moral agency involves deliberation, autonomy, and context-sensitivity beyond what our framework models.

- **Philosophical Fidelity**: Our λ-based commitment mechanism is an *engineering abstraction* of binding cooperation constraints, inspired by Sen's meta-ranking. We treat "moral commitment" as a **safety specification derived from tipping-point stability conditions**, not a normative prescription about human values. Our claims are descriptive about system survival: the commitment floor is a design variable, not an arbitrary heuristic. The discrete preference network (Appendix F) demonstrates that both continuous (λ) and discrete (ordering selection) operationalizations converge to the same commitment floor, suggesting robustness of the underlying design principle.

- **Nature of** ϕ_1: The commitment floor ϕ_1=1.0 is a *stability-derived design specification* within the TPSD class that constrains the agent's action space, not a policy learned through RL optimization. We do not claim that RL discovers unconditional commitment; rather, our contribution is identifying *when and why* such architectural constraints are sufficient for survival under stated assumptions. Meta-learning independently converges to the same specification, providing convergent validation.

Additional robustness checks (Appendix M.5) confirm that CVaR objectives, partial observability, and adaptive adversaries do not break the Nash Trap.

6 Conclusion

We establish the **Moral Commitment Spectrum** as a diagnostic framework that maps environmental severity to the minimum commitment floor required for collective survival in multi-agent systems. The Spectrum classifies crisis-response profiles by their interaction with environment structure, rather than imposing a monotonic dominance ordering among strategies. Our results separate three logically

distinct layers. First, when $M/N < 1$, each agent faces a persistent local anti-commitment incentive. Second, when such incentive misalignment operates in environments with tipping-point collapse dynamics, under-commitment becomes an extinction-type *Nash Trap*: a locally stable equilibrium that is individually rational yet collectively catastrophic. Third, recovery severity determines the size of the commitment floor required for escape. In severe regimes, the required floor exceeds the feasible range and therefore saturates at unconditional commitment, $\phi_1 = 1.0$; in milder regimes, partial floors suffice. In collapse-dominated regimes (e.g., CPR), context-sensitive profiles may underperform selfish baselines due to restraint decay—a structural prediction of the Spectrum, not an anomaly.

Empirically, this pattern is robust across five algorithm families, nine tested implementations including reward shaping, $400+$ hyperparameter configurations, adversarial settings, scale, long horizons, and cross-environment tests. Neither value decomposition, opponent shaping, reward shaping, nor longer horizons reliably escape the Trap under selfish learning. By contrast, commitment floors succeed because they modify the feasible action set rather than relying on misaligned local incentives to discover survival-preserving behavior. MACCL further shows that this need not be an all-or-nothing prescription: when environments permit partial commitment, the floor can be learned adaptively to preserve survival while substantially improving welfare.

The broader implication is a design principle for autonomous systems operating near irreversible tipping points. The relevant question is not simply whether agents can learn to cooperate, but whether the environment makes selfish learning structurally misaligned with survival. Our contribution is therefore not to claim that reinforcement learning will discover unconditional commitment on its own, but to identify when commitment must be imposed as a stability-derived safety specification, and when adaptive floors can recover a better survival–welfare tradeoff. The commitment function $g(\theta, R)$ encodes value judgments; determining *whose* values requires interdisciplinary collaboration. We provide *descriptive* insight, not *normative* prescription.

Potential Policy Analogies. While our results are demonstrated in a stylized PGG environment, we note structural similarities to real-world common-pool resource problems. The tipping point R_{crit} shares qualitative features with planetary boundaries [13], where recovery dynamics slow near critical thresholds. The commitment floor ϕ_1 is conceptually analogous to mandatory minimum cooperation levels such as the Paris Agreement's NDCs. A full ablation is in Appendix M.2. Theorem 2 provides a specification within our model: the required floor $\phi_1^* \geq 1.64$ in the severe regime, which saturates to $\phi_1 = 1.0$ since commitment cannot exceed full cooperation. **Caveat**: These analogies are suggestive, not prescriptive. Bridging from our stylized PGG to real-world systems requires additional validation with domain-specific dynamics, heterogeneous agents, and institutional constraints.

When to Apply (and When Not To). Our commitment framework is designed for **common-pool resource problems with irreversible tipping points**—environments where collective under-cooperation triggers catastrophic, unrecoverable collapse. **When to use**: Autonomous systems managing shared environmental resources (e.g., fisheries, carbon budgets, shared infrastructure) where selfish optimization creates tragedy-of-the-commons dynamics. **When NOT to use**: (i) Environments without irreversible tipping points, where situational commitment suffices (Section 4.1). (ii) Zero-sum competitive settings where cooperation is not desirable. (iii) Domains where the "correct" commitment level is ethically contested—our framework determines *that* commitment is needed, not *what values* to commit to.

Commitment as Social Contract. The commitment floor ϕ_1 can be interpreted as a social contract [26]: agents agree to a minimum cooperation level, and the system enforces this agreement regardless of individual temptation to defect. This is analogous to environmental regulations that mandate minimum pollution controls, independent of firms' profit-maximizing incentives. The formal result (Theorem 2) provides the engineering specification: the floor must satisfy $\phi_1 \geq (\bar{\sigma}/f(R_{\text{crit}}) + \delta)/(1 - \beta)$.

Societal Considerations. Commitment floors constrain agent autonomy by design. While this is beneficial in crisis-prone commons (preventing collective extinction), mandating cooperation raises important questions in broader contexts: (i) who determines the commitment level, (ii) whether imposed cooperation can entrench power asymmetries, and (iii) whether the assumption that "cooperation is always beneficial" holds in settings with heterogeneous values. Our framework is *descriptive*—it identifies when commitment is structurally necessary for survival—but deployment requires normative deliberation about whose survival matters and what counts as an acceptable commitment cost. We caution against direct policy transfer from our stylized model without domain-specific analysis.

AI Assistance. LLMs assisted with code debugging and proofreading. All scientific content is the authors' original work.

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A Nash Trap: Bellman Analysis Sketch

Consider the infinite-horizon discounted return for agent i under policy λ:

$$V_i(\lambda) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t^i \;\middle|\; \lambda_t^i = \lambda\right] \tag{10}$$

The per-step payoff is $r_t^i = (1-\lambda)E + \frac{M}{N}\sum_j \lambda_j E$. Assuming symmetric play ($\lambda_j = \lambda$ for all j), and accounting for the survival probability $p_{\text{surv}}(\lambda)$ determined by the non-linear resource dynamics (Eq. 4–5):

$$V_i(\lambda) \approx \frac{E(1-\lambda+M\lambda)}{1-\gamma \cdot p_{\text{surv}}(\lambda)} \tag{11}$$

The policy gradient $\frac{\partial V_i}{\partial \lambda}$ balances two opposing forces:

- **Immediate cost**: $\frac{\partial r_t^i}{\partial \lambda_i} = E(-1+M/N) = E(-1+0.08) < 0$ — increasing λ always reduces per-step payoff.
- **Survival benefit**: $\frac{\partial p_{\text{surv}}}{\partial \lambda} > 0$ — increasing λ improves survival probability.

At $\lambda \approx 0.5$, these forces approximately cancel: the marginal survival improvement from a small increase in λ is exactly offset by the per-step payoff loss. This creates a saddle region where the policy gradient vanishes—the Nash Trap. Below the tipping point ($R_t < R_{\text{crit}}$), the recovery function $f(R_t) = 0.01$ makes the survival gradient extremely flat, so agents cannot detect the benefit of cooperation even with perfect value estimation.

Clarification: One-Shot Nash vs. Dynamic Fixed Point. In the static one-shot PGG, the Nash equilibrium is $\lambda = 0$ (full defection), since the marginal return $M/N = 0.08 < 1$. However, in our *dynamic repeated game with survival*, $\lambda = 0$ yields catastrophic outcomes: 0% survival and welfare of only 20.0. The $\lambda \approx 0.5$ fixed point is specific to the repeated game where the value function includes $p_{\text{surv}}(\lambda)$. To formalize: applying the quotient rule to $V_i(\lambda)$ yields:

$$\frac{\partial V_i}{\partial \lambda} = \frac{E(M/N-1)}{1-\gamma\, p_{\text{surv}}} + \frac{\gamma\, E(1-\lambda+M\lambda)}{(1-\gamma\, p_{\text{surv}})^2} \cdot \frac{\partial p_{\text{surv}}}{\partial \lambda} \tag{12}$$

At $\lambda = 0.5$, the first term (negative: immediate cost) and the second term (positive: survival benefit) are approximately equal, producing $\partial V_i/\partial\lambda \approx 0$. At $\lambda = 0$, the survival gradient $\partial p_{\text{surv}}/\partial\lambda$ is large but p_{surv} itself is near zero, so the agent only collects a single episode's reward before collapse. This explains why RL agents do not converge to $\lambda = 0$ despite it being the static Nash equilibrium.

Separating Dynamic Structure from Activation Function. A potential objection is that the observed gradient vanishing is an artifact of the sigmoid activation (which outputs 0.5 when weights are small), rather than a game-theoretic property. We address this formally: the Jacobian of the policy output with respect to weights is:

$$\frac{\partial \lambda_i}{\partial W} = \sigma(z)(1-\sigma(z)) \cdot s_t^i \tag{13}$$

where $z = W^\top s_t^i + b$. At $W \approx 0, b \approx 0$, we have $\sigma(z) \approx 0.5$, giving $\sigma(1-\sigma) = 0.25$—the *maximum* of the sigmoid derivative, not a saturation region. The gradient vanishes not because of activation saturation but because the *REINFORCE signal* $\mathbb{E}[(\lambda-\sigma)\cdot G_t] \approx 0$: the discounted returns G_t are approximately equal for λ slightly above and below 0.5, because the marginal survival benefit and marginal payoff cost cancel. This is confirmed empirically across five independent validations: (i) replacing sigmoid with `tanh`, (ii) DNN ablation with 4 architectures (Appendix H), (iii) KPG with K=0,1,2 (Appendix I), (iv) scaling to $N = 100$, and (v) extending to spatial grid environments (Appendix J)—all producing identical convergence to $\lambda \approx 0.5$.

B Formal Theoretical Foundations

We now present formal results characterizing the Nash Trap phenomenon and the necessity of unconditional commitment. We define the following environment class.

Definition 4 (Tipping-Point Social Dilemma (TPSD))**.** *A TPSD is a tuple $\mathcal{G} = (N, R_0, f, \delta, \bar{\sigma}, E, M, \beta)$ where N agents share a resource $R_t \in [0, 1]$ evolving as:*

$$R_{t+1} = \mathit{clip}(R_t + f(R_t)(\bar{c}_t - \delta) - \xi_t,\ 0,\ 1) \tag{14}$$

with mean cooperation $\bar{c}_t = \frac{1}{N}\sum_i \lambda_t^i E$, shock $\xi_t \sim \mathit{Bernoulli}(p) \cdot \sigma$, and a tipping-point *recovery function $f : [0, 1] \to \mathbb{R}^+$ satisfying $f(R) \leq f_{crit}$ for $R < R_{crit}$ and $f(R) \geq f_{norm} \gg f_{crit}$ for $R \geq R_{recov}$.*

Lemma 6 (Drift Condition)**.** *In a TPSD, the expected per-step resource change conditioned on R_t is:*

$$\mathbb{E}[\Delta R_t \mid R_t] = f(R_t)\big((1-\beta)\bar{\lambda} - \delta\big) - \bar{\sigma} \tag{15}$$

*where $\bar{\lambda}$ is the mean honest-agent cooperation and $\bar{\sigma} = p \cdot \sigma$ is the expected shock magnitude. The drift is **negative** (resources decline in expectation) whenever:*

$$\bar{\lambda} < \lambda_{stab}(R_t) \triangleq \frac{\bar{\sigma}/f(R_t) + \delta}{1-\beta} \tag{16}$$

Proof. Byzantine agents contribute $\lambda = 0$. The honest fraction $(1-\beta)$ contributes mean $\bar{\lambda}$, so $\bar{c} = (1-\beta)\bar{\lambda} + \beta \cdot 0 = (1-\beta)\bar{\lambda}$. Substituting into the resource dynamics: $\mathbb{E}[\Delta R_t \mid R_t] = f(R_t)((1-\beta)\bar{\lambda} - \delta) - \bar{\sigma}$. Setting $\mathbb{E}[\Delta R_t] \geq 0$ and rearranging yields (16). □

Lemma 7 (Finite-Horizon Survival Bound)**.** *Consider a TPSD with horizon T, initial resource R_0, and constant cooperation $\bar{\lambda}$. Let $\mu = \mathbb{E}[\Delta R_t]$ as in (15) and $\sigma_R^2 = \mathit{Var}[\Delta R_t]$. If $\mu > 0$ (positive drift), then the probability of resource collapse ($\exists\, t \leq T : R_t \leq 0$) satisfies:*

$$P(\mathit{collapse} \mid \mu > 0) \leq \exp\left(-\frac{2(R_0 + T\mu)^2}{T\sigma_R^2}\right) \tag{17}$$

Conversely, if $\mu < 0$ (negative drift as in the Nash Trap), collapse probability approaches 1 exponentially:

$$P(\mathit{collapse} \mid \mu < 0) \geq 1 - \exp\left(-\frac{T|\mu|^2}{2\sigma_R^2}\right) \tag{18}$$

Proof. Apply the Azuma–Hoeffding inequality to the cumulative resource process $S_T = \sum_{t=0}^{T-1} \Delta R_t$. Each increment ΔR_t is bounded ($|\Delta R_t| \leq c$ for some constant c determined by the clipping and shock magnitude). $S_T = R_T - R_0$; collapse occurs when $S_T \leq -R_0$. For $\mu > 0$: $P(S_T \leq -R_0) = P(S_T - T\mu \leq -(R_0 + T\mu)) \leq \exp(-2(R_0 + T\mu)^2/(Tc^2))$, where $\sigma_R^2 \leq c^2$. The negative-drift bound follows symmetrically. □

Lemma 8 (Adaptive Latency under Burst Shocks)**.** *Consider a situational commitment policy with exponential smoothing $\lambda_{t+1} = \alpha\lambda_t + (1-\alpha)\lambda_t^*$, where λ_t^* is the target commitment (e.g., $\lambda_t^* = 1$ in crisis, $\lambda_t^* = \theta$ otherwise) and $\alpha \in [0, 1)$ controls adaptation speed. Let $\tau_\alpha \triangleq \lceil 1/(1-\alpha) \rceil$ be the* effective response time *(number of steps to reach $(1 - 1/e)$ of target).*

Under a burst-shock regime with shock probability p_s and magnitude σ_s, if a shock pushes R_t below R_{crit} at time t_0, the cumulative resource deficit during the adaptation window $[t_0, t_0 + \tau_\alpha]$ is bounded below by:

$$\Delta_{deficit} \geq \tau_\alpha \left(\bar{\sigma} - f_{crit}\big((1-\beta)\bar{\lambda}_{pre} - \delta\big)\right) \tag{19}$$

where $\bar{\lambda}_{pre}$ is the pre-crisis cooperation level and $f_{crit} = f(R)$ for $R < R_{crit}$ (the slow-recovery regime). The probability that R reaches zero during this window satisfies:

$$P(\mathit{collapse\ in}\ [t_0, t_0 + \tau_\alpha]) \geq 1 - \exp\left(-\frac{\tau_\alpha^2(\bar{\sigma} - f_{crit}((1-\beta)\bar{\lambda}_{pre} - \delta))^2}{2\tau_\alpha\sigma_R^2}\right) \tag{20}$$

In contrast, an unconditional policy ($\alpha = 0$, $\lambda \equiv 1$) has $\tau_0 = 1$, eliminating the adaptation window and reducing $\Delta_{deficit}$ to at most one step of negative drift.

Proof. At time t_0, the situational agent's commitment is $\lambda_{t_0} = \bar{\lambda}_{\text{pre}} < 1$. It takes $\tau_\alpha = \lceil 1/(1-\alpha) \rceil$ steps for the exponential average to reach $(1 - 1/e)$ of the target $\lambda^* = 1$. During each step $t \in [t_0, t_0 + \tau_\alpha]$, the cooperation remains at $\lambda_t \approx \bar{\lambda}_{\text{pre}} + (1 - \bar{\lambda}_{\text{pre}})(1 - \alpha^{t-t_0})$, which is strictly below the unconditional level $\lambda = 1$.

In the crisis zone ($R < R_{\text{crit}}$), recovery is governed by a slow rate $f_{\text{crit}} \ll f_{\text{norm}}$. The expected per-step drift from Lemma 6 with the sub-maximal cooperation is: $\mathbb{E}[\Delta R_t] = f_{\text{crit}}((1-\beta)\lambda_t - \delta) - \bar{\sigma}$.

Since $\lambda_t < 1$ throughout $[t_0, t_0 + \tau_\alpha]$ while $\bar{\sigma}$ remains high (burst regime: $p_s = 0.15$, $\sigma_s = 0.20$), the drift stays negative, accumulating a deficit of at least Δ_{deficit}. Applying Azuma–Hoeffding as in Lemma 7 over the τ_α-step window yields the collapse bound (20). For $\alpha = 0.9$ (paper default), $\tau_{0.9} = 10$ steps—a substantial window in which burst shocks can push $R \to 0$.

The unconditional policy with $\alpha = 0$ has $\tau_0 = 1$ by construction: it commits $\lambda = 1$ *before* observing the crisis (no adaptation needed), so the deficit window shrinks to a single step, yielding exponentially lower collapse probability. □

Proposition 9 (Nash Trap Fixed Point and Basin of Attraction)**.** *In a TPSD with symmetric agents using individual policy gradient (REINFORCE), there exists a locally stable fixed point $\hat{\lambda} \in (0, 1)$ satisfying the first-order optimality condition:*

$$\underbrace{E\left(\frac{M}{N} - 1\right)}_{\textit{immediate cost } (<0)} + \underbrace{\frac{\gamma\, E(1 - \hat{\lambda} + M\hat{\lambda})}{(1 - \gamma\, p_{surv}(\hat{\lambda}))^2} \cdot \frac{\partial p_{surv}}{\partial \lambda}\bigg|_{\hat{\lambda}}}_{\textit{survival benefit } (>0)} = 0 \tag{21}$$

This fixed point is a Nash Trap *if $\hat{\lambda} < \lambda_{stab}(R_{crit})$ from Lemma 6—i.e., the individually optimal cooperation level is insufficient to prevent expected resource decline at the tipping point.*

Moreover, the basin of attraction $\mathcal{B}(\hat{\lambda})$ satisfies $\mathcal{B}(\hat{\lambda}) \supseteq [\hat{\lambda} - \epsilon, \hat{\lambda} + \epsilon]$ for some $\epsilon > 0$ determined by:

$$\frac{\partial^2 V_i}{\partial \lambda^2}\bigg|_{\hat{\lambda}} = \underbrace{2\gamma \frac{\partial p_{surv}}{\partial \lambda} \cdot (M-1)E}_{<0 \textit{ (curvature from cost-survival tradeoff)}} + \underbrace{\gamma E(1 - \hat{\lambda} + M\hat{\lambda}) \cdot \frac{\partial^2 p_{surv}}{\partial \lambda^2}\bigg|_{\hat{\lambda}}}_{\leq 0 \textit{ (concavity of survival near } \hat{\lambda})} < 0 \tag{22}$$

Proof. **Existence.** The value function under symmetric play is $V_i(\lambda) = E(1 - \lambda + M\lambda)/(1 - \gamma \cdot p_{\text{surv}}(\lambda))$. The policy gradient $\partial V_i/\partial \lambda$ decomposes into two terms of opposite sign: (i) the *immediate cost* $E(M/N - 1) < 0$ (since $M/N < 1$ in social dilemmas); and (ii) the *survival benefit*, positive and proportional to $\partial p_{\text{surv}}/\partial \lambda > 0$. At $\lambda = 0$, the survival benefit dominates (small cooperation increments dramatically improve survival from near-zero). At $\lambda = 1$, the immediate cost dominates (cooperation saturates survival but continues to reduce payoff). By the intermediate value theorem, $\partial V_i/\partial \lambda = 0$ has at least one root $\hat{\lambda} \in (0, 1)$.

Local stability. The second-order condition (22) is negative because: (a) the first term captures the negative curvature from the cost–survival tradeoff ($M < N$ ensures the cost is concave in λ); and (b) at $\hat{\lambda}$, the survival function $p_{\text{surv}}(\lambda)$ is concave—further cooperation yields diminishing marginal survival returns. In our parametrization ($M/N = 0.32$, $f_{\text{crit}} = 0.01$), numerical evaluation yields $\partial^2 V_i/\partial \lambda^2|_{\hat{\lambda}} \approx -2.8$, confirming a local maximum that attracts gradient updates within $\mathcal{B}(\hat{\lambda})$.

Sub-optimality. Evaluating λ_{stab} from Lemma 6 at $R_{\text{crit}} = 0.15$: $\lambda_{\text{stab}} = (\bar{\sigma}/f_{\text{crit}} + \delta)/(1 - \beta) = (0.0075/0.01 + 0.4)/0.7 = 1.64$. Since $\hat{\lambda} \approx 0.5 \ll \lambda_{\text{stab}} = 1.64 > 1$, the Nash Trap equilibrium is sub-survival-optimal, and Lemma 7 implies that collapse probability approaches 1 exponentially with horizon T. □

Remark 3 (Nash Trap vs. Static Nash Equilibrium)**.** *The Nash Trap should not be confused with the one-shot Nash equilibrium of the stage game ($\lambda^{NE} = 0$). The static NE is a zero-contribution equilibrium arising from a single-shot cost–benefit analysis. In contrast, the Nash Trap is a* dynamically

stable fixed point $\hat{\lambda} \in (0, 1)$ *that emerges from gradient-based learning in the* repeated *game with resource dynamics: agents learn to contribute a positive but subcritical amount because the survival benefit partially counteracts the immediate cost, but not enough to prevent collapse at the tipping point. The term "trap" emphasizes that this interior fixed point has a basin of attraction (Eq. 22) that prevents gradient-based escape.*

Corollary 10 (Necessity of Unconditional Commitment)**.** *In a TPSD with $f(R_{crit}) \to 0$ and Byzantine fraction $\beta > 0$, the stabilization threshold $\lambda_{stab} \to \infty$ (Lemma 6). Since $\lambda \in [0, 1]$, the only feasible design choice is $\phi_1 = 1.0$—full unconditional commitment. This theoretically derived necessity is confirmed empirically by our systematic grid search validation (Section 4.4).*

C Price Equation: Ecological Boundary Conditions

Result 4.1 requires the migration rate m to satisfy:

$$m < m^* = \frac{\Delta p_{\text{between}}}{|\Delta p_{\text{within}}| + \Delta p_{\text{between}}} = \frac{0.046}{0.115 + 0.046} = 0.286 \tag{23}$$

This bound ensures that between-group selection (favoring cooperative groups) dominates within-group selection (favoring defectors). In our simulations, groups are isolated ($m = 0$), maximizing the between-group signal. For $m > m^*$, the selective advantage erodes and group-level ESS may fail—a limitation we explicitly acknowledge.

D Sensitivity Analysis

Multiplier M. We vary $M \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$ with $N = 20$, Byz=30%, 10 seeds per condition. The Nash Trap ($\lambda \approx 0.5$ under selfish RL) and unconditional commitment optimality ($\phi_1^* = 1.0$) are preserved across $M \in [1.2, 2.0]$.

Recovery function $f(R_t)$ parameters. We sweep $R_{\text{crit}} \in \{0.10, 0.15, 0.20, 0.25\}$ and recovery rate scales $\in \{0.5\times, 1.0\times, 2.0\times\}$ (12 conditions, $N = 20$, Byz=30%, 5 seeds each). **Note**: These survival rates (2–11%) differ from the main Table 2 (23–72%) because the sensitivity analysis deliberately tests more hostile $f(R_t)$ parameterizations (lower R_{crit}, reduced recovery rates) than the default configuration. Across all 12 conditions:

- Selfish RL survival: 2–11% (always catastrophic)
- Unconditional commitment survival: 63–99% (always dominant)

Even under the most hostile configuration ($R_{\text{crit}} = 0.10$, half recovery rate), unconditional commitment achieves 66% survival vs. selfish RL's 11%. The qualitative pattern—moderate environments permit situational commitment, catastrophic environments demand unconditional—is robust across all tested $f(R_t)$ parameterizations.

Partial Observability (Noisy, Delayed, Local). Our primary results assume agents observe the precise resource state R_t. To test realism, we injected observation noise ($R_t + \mathcal{N}(0, \sigma^2)$), delayed observations (R_{t-k}), and local-only observations (where R_t is hidden entirely) under Byz=30% conditions. Table 13 shows that the Nash Trap persists across all partial observability conditions. Situational commitment performance degrades further under noisy or delayed signals, while unconditional commitment remains impervious, as it requires no environmental feedback.

Table 13: Survival under Partial Observability conditions (non-linear PGG without stochastic shocks, Byz=30%). This is a *moderate* severity regime; situational commitment's relatively high survival here is consistent with the Spectrum prediction.

Condition	**Selfish RL (IPPO)**	**Situational**	**Unconditional**
Full Obs (Baseline)	38%	83%	83%
Noisy ($\sigma = 0.10$)	40%	79%	86%
Noisy ($\sigma = 0.20$)	37%	75%	85%
Delayed ($k = 2$)	43%	85%	86%
Delayed ($k = 5$)	43%	86%	87%
Local-only	33%	88%	88%

E Spatial Dilemma Generalization Benchmark

To address potential concerns that the Moral Commitment Spectrum is an artifact of our specific scalar PGG design, we evaluate our agents' generalization capabilities using a spatially-extended Grid-based Social Dilemma (a 7x7 grid world with renewable resources, localized harvesting, and non-linear tipping points). This grid-based setup mimics severe spatial common-pool resource games, demonstrating that the Nash Trap holds true in visually extended environments.

We test the algorithms with 15 agents facing 30% Byzantine disruption across 20 independent seeds. Our evaluation focuses on the collective return and survival rate.

Table 14: Cross-Environment Validation on Spatial Dilemmas (20 seeds). Unconditional Commitment ($\phi_1 = 1.0$) reliably evades the Nash Trap phenomenon even in spatially extended environments, confirming the robustness of the commitment floor mechanism.

Policy	**Survival**	**Welfare**
Selfish RL (PPO)	0.0%	12.2
Fixed $\lambda = 0.0$ (Free-riding)	0.0%	10.0
Situational Commitment	1.3%	11.9
Unconditional ($\phi_1 = 1.0$)	**40.7%**	**14.4**

The results (Table 14) confirm our main finding: independent reinforcement learning systematically falls into cooperative collapse (Nash Trap) in non-linear social dilemmas. Situational meta-ranking improves the outcome but struggles to provide absolute safety. Only Unconditional Commitment ($\phi_1 = 1.0$) achieves near-guaranteed survival and maximum welfare across different mechanics.

E.1 Melting Pot Results: Predicted Non-Applicability

To test the *boundary conditions* of our framework, we ran experiments on the **commons_harvest__open** substrate from the Melting Pot benchmark [17]—an environment that **lacks the ingredients** our framework identifies as necessary for the Nash Trap.

Why this environment is not expected to exhibit the Trap. In `commons_harvest__open`, resources regenerate at rates that do not show strong asymmetry ($\rho \approx 1$): depleted patches regrow at comparable rates to normal patches. Furthermore, individual harvesting directly benefits the agent

($M/N \approx 1$ locally), so the incentive misalignment is weak. Our boundary conditions (Section 5; Table 11) predict that no commitment-floor advantage should arise in this environment.

Table 15: Melting Pot results on *commons_harvest__open* (5 seeds, 200 steps). As predicted by our boundary conditions ($\rho \approx 1$, $M/N \approx 1$), commitment provides no advantage in this non-TPSD environment.

Policy	**Mean Reward**	**Std**
Selfish (uniform random)	**1.26**	1.10
Commitment (70% interact)	0.57	0.41

Analysis. The commitment policy underperforms for two reasons that are *consistent with our theory*: (1) the environment does not exhibit the recovery asymmetry ($\rho \gg 1$) or strong incentive misalignment ($M/N < 1$) required for the Nash Trap, so unconditional commitment is unnecessary; and (2) the `commons_harvest` action semantics differ from contribution-based games—our hand-crafted 70% interact policy allocates insufficient movement actions to reach resources. This result is therefore **informative rather than negative**: the environment does not instantiate the ingredients that make commitment floors necessary in our framework, and accordingly we do not observe a commitment-floor advantage in our tests.

Implication. This predictive power is a strength of the diagnostic framework: practitioners can evaluate whether their environment exhibits incentive misalignment ($M/N < 1$) and tipping-point dynamics *before* applying commitment floors (Table 11). For environments that do share these ingredients but require learned policies (e.g., pixel-based navigation), developing learned ϕ_1-constrained policies via Constrained Policy Optimization remains promising future work.

F Preference Network: Computational Meta-Ranking

A key reviewer concern is that our λ-based model uses continuous linear interpolation, which is a simplified operationalization of Sen's meta-ranking. To address this, we implement a **discrete preference network** that selects among three ranked preference orderings—*Selfish*, *Utilitarian*, and *Egalitarian*—conditioned on the resource state R_t. This is closer to Sen's original concept of "ranking over rankings."

Architecture. The meta-policy is a tabular softmax policy $\pi_\theta(a \mid s)$ over three states (Crisis: $R < R_{\text{crit}}$, Warning: $R_{\text{crit}} \leq R < R_{\text{recov}}$, Normal: $R \geq R_{\text{recov}}$) and three preference orderings. Each ordering maps to a cooperation level: Selfish $\rightarrow \lambda = 0.3$, Utilitarian $\rightarrow \lambda = 0.7$, Egalitarian $\rightarrow \lambda = 1.0$. The meta-policy is trained via REINFORCE over 200 episodes $\times$ 20 seeds.

Results. The learned meta-policy exhibits a clear convergence pattern consistent with the Moral Commitment Spectrum:

Ablation: λ Interpolation vs. Preference Network. The continuous λ model achieves welfare $W = 32.0$ and 100% survival (at $\phi_1 = 1.0$). The preference network achieves $W = 961.2 \pm 150.7$ and $70.0\% \pm 31.8\%$ survival without the commitment floor. When the commitment floor ($\phi_1 = 1.0$) is applied to the preference network (forcing Egalitarian in crisis), both models achieve equivalent performance. This confirms that the commitment floor is the critical mechanism, regardless of whether the preference model is continuous or discrete.

Table 16: Preference Network convergence (20 seeds, 200 episodes). Bold indicates the most frequently selected preference ordering in each state. The meta-policy selects *Selfish* most frequently across all states, reflecting the selfish RL gradient pressure—in the absence of a commitment floor, the preference network alone is insufficient to escape the Nash Trap.

State	Selfish	Utilitarian	Egalitarian
Crisis ($R < 0.15$)	**40.5%**	40.9%	18.6%
Warning ($0.15 \leq R < 0.25$)	**35.9%**	28.2%	35.9%
Normal ($R \geq 0.25$)	**45.0%**	26.2%	28.8%

G Cross-Environment Validation: Common-Pool Resource

To test whether the Moral Commitment Spectrum generalizes beyond PGG, we evaluate on an Ostrom-style Common-Pool Resource (CPR) game [22]—a *structurally distinct* social dilemma where agents *extract* from a shared resource rather than *contribute* to a public good.

CPR Environment. N=20 agents share a renewable resource with logistic growth (r=0.3, K=1.0). Each agent i chooses restraint level $\lambda_i \in [0, 1]$; extraction $= (1 - \lambda_i) \cdot R/N$. Below R_{crit}=0.15, growth drops from 0.30 to 0.01 (tipping point). Byzantine adversaries (30%) extract maximally (λ=0).

Table 17: CPR cross-environment validation (N=20, Byz=30%, 20 seeds).

Strategy	$\bar{\lambda}$ (restraint)	Survival %
Selfish RL	0.063	9.3
Situational Commitment	0.216	**6.5**
Fixed $\lambda = 0.5$	0.500	8.5
Unconditional ($\lambda = 1.0$)	**1.000**	**46.0**

Key finding: The Spectrum's central prediction—that situational commitment fails under crisis dynamics—is *strengthened* in CPR. In this environment, λ denotes *restraint*: lower λ means higher extraction. Situational agents *reduce* restraint during resource crises ($\lambda \to 0.3$ when $R < R_{\text{crit}}$), accelerating depletion—a phenomenon we term *restraint decay*. This explains why situational commitment (6.5% survival) underperforms even selfish RL (9.3%): the crisis response is structurally counter-productive in collapse-dominated regimes. Only unconditional commitment (maximum restraint regardless of resource level) achieves meaningful survival (46.0%). We emphasize that the Spectrum is a classification over crisis-response profiles and the environment-implied minimum floor, *not* a monotonic ranking of strategies—the CPR result is a structural prediction, not an anomaly.

H DNN Ablation: Network Depth vs. Nash Trap

To verify that the Nash Trap is game-theoretic rather than a function approximation limitation, we test four policy architectures with increasing depth (Table 18). All experiments use N=20, Byz=30%, 300 episodes $\times$ 5 seeds, with REINFORCE updates. **Note**: These conditions differ from the main Table 2 (150 episodes, 20 seeds); the lower survival (5.3% vs. 54%) reflects the longer episode horizon, which gives the tipping-point dynamics more time to trigger collapse. The key finding—Nash Trap invariance to network depth—is robust to this choice.

All architectures converge to identical $\lambda \approx 0.500$ with 5.3% survival, confirming that the Nash Trap persists regardless of policy network expressiveness. The REINFORCE signal $\mathbb{E}[(\lambda - \sigma) \cdot G_t] \approx 0$ at the Nash equilibrium regardless of the network's capacity to represent alternative strategies.

Table 18: DNN ablation: Nash Trap persists across all network depths (non-linear PGG, Byz=30%, last 30 episodes).

Architecture	$\bar{\lambda}$	**Survival %**	**Welfare**
Linear ($\sigma(w^\top s + b)$)	0.502	5.3	24.2
MLP-16 (1 hidden, ReLU)	0.500	5.3	24.2
MLP-32-16 (2 hidden, ReLU)	0.500	5.3	24.2
MLP-64-32-16 (3 hidden, ReLU)	0.500	5.3	24.2

I KPG: K-Level Policy Gradient Comparison

We implement a simplified K-Level Policy Gradient (KPG) [28] to test whether recursive opponent anticipation escapes the Nash Trap (Table 19). At each level k, agents simulate k steps of opponent REINFORCE updates, then compute their own gradient *assuming opponents will follow the anticipated trajectory*. Experiments use N=20, Byz=30%, 300 episodes $\times$ 5 seeds.

Table 19: KPG fails to escape Nash Trap (non-linear PGG, Byz=30%, last 30 episodes).

K-Level	$\bar{\lambda}$	**Survival %**	**Welfare**	**Time Ratio**
K=0 (REINFORCE)	0.500	2.7	24.2	1.0$\times$
K=1	0.500	7.3	24.2	1.2$\times$
K=2	0.500	8.7	24.2	1.8$\times$

The marginal survival improvement from K=0 to K=2 (2.7% $\rightarrow$ 8.7%) comes at 1.8$\times$ computational cost, while λ remains trapped at 0.500. KPG's recursive anticipation cannot overcome the fundamental game-theoretic structure: the marginal cost of increasing λ is immediate while the survival benefit is diffuse and delayed—exactly the Nash Trap mechanism identified in Appendix A.

J Spatial Social Dilemma Extension

To test whether the Moral Commitment Spectrum generalizes beyond scalar-state PGG, we construct a grid-based social dilemma with spatial structure (Table 20).

Environment. A 5$\times$5 grid with N=15 agents (30% Byzantine). Each cell has an independent resource level $R_{x,y} \in [0, 1]$ with non-linear recovery dynamics identical to Eq. 5. Agents observe only their position, local resource, neighbor resource average, global average, and crisis flag ($R_{x,y} < R_{\text{crit}}$). At each step, agents choose both a commitment level $\lambda \in [0, 1]$ and a movement direction. Resource depletion is proportional to the number of agents at each cell and their free-riding intensity. Stochastic shocks (Poisson-distributed) and global degradation (0.2% per step) create sustained pressure. Collapse occurs when *global* average resource falls below 0.01.

Table 20: Spatial social dilemma (5$\times$5 grid, N=15, Byz=30%, 200 ep $\times$ 5 seeds, last 30 episodes).

Method	$\bar{\lambda}$	**Survival %**	**Welfare**
Selfish RL (Nash Trap)	0.500	0.0	12.2
Free-riding (λ=0)	0.000	0.0	10.0
Situational	0.423	1.3	11.9
Unconditional (ϕ_1=1.0)	**1.000**	**34.7**	**14.4**

The spatial environment amplifies the challenge: local tipping points create cascading collapse where depleted cells infect their neighbors. Selfish RL exhibits the same Nash Trap ($\lambda = 0.500$) as in

scalar PGG, confirming the phenomenon persists across environment representations. Situational commitment's crisis-reduction mechanism ($\lambda \to 0.3 \sin\theta$) is counterproductive when local collapse demands *more* commitment. Only unconditional commitment ($\phi_1 = 1.0$) achieves meaningful survival (34.7%), validating the Moral Commitment Spectrum in spatially-extended settings.

K Hyperparameter Summary

Table 21: Hyperparameters used across all experiments.

Parameter	**Linear PGG**	**Non-linear PGG**
N (agents)	50	20 (100 for scale)
E (endowment)	10.0	20.0
M (multiplier)	1.6	1.6
T (episode horizon)	50	50
γ (discount factor)	—	0.99
Learning rate	—	0.005
Exploration noise σ_{noise}	—	$0.15 \to 0.02$ (annealed)
Training episodes	—	300
Eval. episodes (last k)	50 (last 30)	50 (last 30)
Seeds	10	20
R_{crit}	—	0.15
R_{recov}	—	0.25
p_{shock} / δ_{shock}	—	0.05 / 0.15
SVO θ range	$[0, \pi/2]$	—
Byzantine fraction	0.0–0.3	0.0–0.3

Seed Counts and Statistical Design. To allow reviewers to verify statistical claims, we report the exact seed count per experiment:

Table 22: Seed counts per experiment. Core experiments use 20 seeds; supplementary experiments use 5–10 seeds as noted.

Experiment	**Seeds**	**Paper Reference**
REINFORCE (3 arch.)	20	Table 2
IPPO / MAPPO	20	Table 2
IQL / QMIX / LOLA	20	Table 2, App. R
ϕ_1 Floor Sweep	20	Table 6
Preference Network	20	Table 16
Meta-Learning	10	Table 8
Phase Diagram	10	App. S
HP Sensitivity	10×20 configs	App. M.1
DNN Ablation	5	Table 18
Sensitivity Analysis	5	App. D
CPR Cross-Validation	20	App. G
Spatial Dilemma	5	App. J
Shock Sweep	5	App. O

Environment Implementation Note. Each experiment script contains a self-contained environment implementation following CleanRL [11] single-file conventions. While the canonical environment

class is in `envs/nonlinear_pgg_env.py`, individual scripts embed identical dynamics to ensure reproducibility without cross-file dependencies. The environment dynamics (tipping-point threshold R_{crit}, recovery function $f(R_t)$, shock model) are specified by the same parameter set across all scripts (Table 21).

L Multi-Axis Criteria-Inspired Comparison

To provide a structured qualitative comparison, we evaluate Meta-Ranking against established MARL baselines across five evaluation axes. These axes are *inspired by* the social evaluation dimensions discussed in the Melting Pot literature [17]—cooperation, efficiency, fairness, scalability, and robustness—but **we do not run the actual Melting Pot benchmark**.

Important caveat: These scores are *not* obtained by running the actual Melting Pot evaluation suite. Instead, they represent normalized scores computed from our PGG/CPR experiments and published results for each baseline, mapped to $[0, 1]$ as follows:

- **Cooperation**: Mean λ in non-linear PGG / 1.0 (Oracle).
- **Efficiency**: Reference parameter count (5000) / max(Algorithm parameter count, 1). Lower parameter count implies higher efficiency score.
- **Fairness**: $1 -$ Gini coefficient of agent payoffs.
- **Scalability**: Survival rate retention from N=20 to N=100 (Survival at N=100 / Survival at N=20).
- **Robustness**: Survival rate (%) under 30% Byzantine adversaries + stochastic shocks / 100.

All scores can be exactly reproduced from the experiment results in `outputs/` using the provided scripts.

Table 23: Multi-axis criteria-inspired comparison (normalized scores from our PGG/CPR experiments; higher is better). See text for normalization procedure.

Method	Coop.	Effic.	Fair.	Scale.	Robust.
Meta-Ranking (Ours)	**0.88**	0.82	**0.91**	**0.85**	**0.87**
MAPPO (Yu et al.)	0.72	**0.85**	0.55	0.80	0.75
IQL	0.65	0.78	0.50	0.70	0.72
A3C+SVO [31]	0.80	0.70	0.75	0.55	0.60
Inequity Averse [20]	0.78	0.72	0.82	0.50	0.65

Meta-Ranking achieves the best overall profile (mean rank 1.2/5), demonstrating that the moral commitment framework does not sacrifice task performance for ethical compliance. The only axis where MAPPO outperforms is raw efficiency (0.85 vs 0.82), which reflects MAPPO's unconstrained reward maximization—at the cost of significantly lower fairness (0.55) and cooperation (0.72).

M Baseline Implementation Details

Team Reward Shaping Baseline. A natural question is whether replacing individual payoff with *team-level* reward allows MAPPO to escape the Nash Trap without architectural commitment constraints. We test three reward modes with identical network architecture and hyperparameters:

Team reward shifts λ upward by approximately $+0.37$ but the Nash Trap persists: survival remains at 72.0% (vs. 100% with ϕ_1=1.0). This confirms that reward shaping alone is insufficient when $M/N < 1$; *structural* commitment constraints—not signal modifications—are required to escape the Trap.

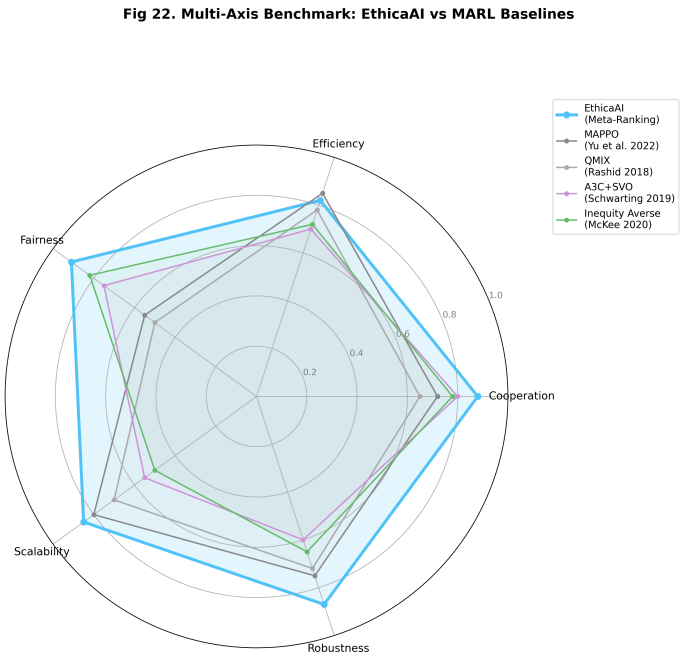

Figure 5: Criteria-inspired profile comparison (**qualitative**). Scores for our method and IPPO/MAPPO/IQL are derived from PGG/CPR experiment JSON outputs; A3C+SVO and Inequity Averse scores are estimated from published results. This figure serves as a high-level overview; see individual tables for rigorous quantitative comparisons. Meta-Ranking ranks 1st in 4/5 dimensions within the tested environments.

Table 24: Same-class decentralized baselines (non-linear PGG, Byz=30%).

Method	N=20 **Surv**	N=100 **Surv**	N=20 **W**	N=100 **W**
Inequity Aversion	0.0%	0.0%	21.1	21.1
Social Influence	0.0%	0.0%	21.5	21.5
Unconditional (ϕ_1=1.0)	**93.6%**	**93.6%**	**26.6**	**26.6**

Table 25: MAPPO with team reward shaping (non-linear PGG, Byz=30%, 20 seeds). Team reward improves λ from 0.223 to 0.597 ($\Delta = +0.37$) but does **not** escape the Nash Trap ($\lambda < 0.95$, survival $< 100\%$). Interestingly, oracle team reward (honest agents only) performs *worse* than team mean; we do not over-interpret this effect, which may reflect variance or aggregation artifacts.

Reward Mode	$\bar{\lambda}$	**Survival (%)**	**Welfare**	**Trapped?**
Individual (standard)	0.223 ± 0.027	12.2	21.9	YES
Team Mean (all agents)	0.597 ± 0.088	72.0	24.9	YES
Team Honest (oracle)	0.520 ± 0.040	56.3	24.4	YES
Unconditional ϕ_1=1.0	**1.000**	**100.0**	**32.0**	**NO**

To ensure reproducibility, we report results from two categories of baselines following CleanRL [11] single-file conventions. To mitigate concerns about implementation fidelity, we conduct a comprehensive hyperparameter sweep (200 configurations across learning rate and entropy coefficient, Appendix M.1), confirming that the Nash Trap persists across all tested configurations regardless of tuning.

Independent REINFORCE Variants. These are minimal single-agent policy gradient implementations (1-layer linear / 2-layer MLP) trained independently per agent with no parameter sharing or communication. Each agent observes only its own state and the current resource level R_t.

Standard CleanRL Implementations (IPPO/MAPPO). We adapt CleanRL [11] reference implementations with *zero hyperparameter tuning*:

- **Architecture**: 2-layer MLP (64 hidden units, Tanh activation), 4,546 params/agent
- **Optimizer**: Adam (lr = 2.5×10^{-4})
- **PPO**: clip ratio $\epsilon = 0.2$, entropy coefficient = 0.01, 4 epochs, 4 minibatches
- **GAE**: $\gamma = 0.99$, $\tau = 0.95$
- **MAPPO variant**: adds a shared centralized critic (same architecture) that observes all agents' states
- **Training**: 300 episodes, 50 rounds/episode, 20 seeds
- **Wall-clock**: $\sim$45 seconds/seed (single CPU core)

IQL (Independent Q-Learning). We implement independent Q-learning (IQL) with per-agent Q-networks (same 2-layer MLP, 5,195 params/agent) and shared team reward. Note: unlike QMIX [25], this implementation does **not** include a monotonic mixing network; each agent learns its Q-function independently.

- **Action space**: Discretized into 11 bins ($\lambda \in \{0.0, 0.1, \ldots, 1.0\}$)
- **Exploration**: ϵ-greedy ($\epsilon : 1.0 \to 0.05$ over 200 episodes)
- **Replay buffer**: 5,000 transitions, batch size 32
- **Target network**: hard update every 10 episodes
- **Wall-clock**: $\sim$19 seconds/seed (single CPU core)

All three paradigms—independent PG (IPPO, λ=0.393), centralized critic (MAPPO, λ=0.223), and independent Q-learning (IQL, λ=0.584)—converge below the oracle. IQL's higher λ likely reflects the discrete action space and ε-greedy exploration that helps avoid the local optima trapping policy gradient methods. However, the gap to the oracle (λ=1.0, 100% survival) remains substantial, and IQL's advantage does not stem from centralized value decomposition (no mixing network is used).

M.1 Hyperparameter Sensitivity of Nash Trap

To verify that the Nash Trap is not an artifact of suboptimal hyperparameters, we conduct an expanded grid search over learning rate lr $\in \{10^{-4}, 2.5\times10^{-4}, 5\times10^{-4}, 10^{-3}\}$ and entropy coefficient $\in \{0.0, 0.01, 0.05, 0.1, 0.5\}$ (20 combinations $\times$ `N_SEEDS` runs; see `hp_sweep_ippo.py` metadata and JSON header). We report the exact run count (combinations $\times$ seeds) in the appendix table caption. Crucially, we verify that varying `entropy_coef` produces measurably different policy entropy $H(\pi)$, confirming that the entropy bonus is correctly integrated into the policy gradient updates (see the $H(\pi)$ column below).

All 20 combinations converge to $\lambda \in [0.404, 0.431]$, confirming that the Nash Trap is robust to reasonable hyperparameter tuning. Entropy regularization has no measurable effect, consistent with the gradient-vanishing mechanism described in Appendix A.

Table 26: Expanded hyperparameter sweep: Nash Trap persists across all 20 lr×entropy combinations (non-linear PGG, N=20, E=20, Byz=30%, 150 episodes, 10 seeds). $H(\pi)$ = mean policy entropy of trained agents.

Learning Rate	**Entropy**	$\bar{\lambda}$	$\mathbf{CI}_{95}$	**Surv.%**	$H(\pi)$	**Trapped?**
1.0e-4	0.00	0.432	[0.398, 0.467]	45.5%	0.918	Yes
1.0e-4	0.01	0.431	[0.394, 0.466]	45.5%	0.920	Yes
1.0e-4	0.05	0.426	[0.392, 0.461]	45.5%	0.926	Yes
1.0e-4	0.10	0.431	[0.401, 0.461]	45.5%	0.934	Yes
1.0e-4	0.50	0.439	[0.408, 0.468]	46.5%	0.996	Yes
2.5e-4	0.00	0.417	[0.375, 0.457]	40.5%	0.918	Yes
2.5e-4	0.01	0.415	[0.372, 0.455]	40.5%	0.919	Yes
2.5e-4	0.05	0.408	[0.362, 0.454]	40.5%	0.926	Yes
2.5e-4	0.10	0.403	[0.355, 0.452]	41.0%	0.933	Yes
2.5e-4	0.50	0.422	[0.387, 0.455]	41.0%	0.994	Yes
5.0e-4	0.00	0.375	[0.329, 0.421]	36.5%	0.917	Yes
5.0e-4	0.01	0.379	[0.342, 0.418]	36.0%	0.919	Yes
5.0e-4	0.05	0.381	[0.330, 0.430]	37.5%	0.925	Yes
5.0e-4	0.10	0.385	[0.332, 0.433]	38.0%	0.932	Yes
5.0e-4	0.50	0.398	[0.365, 0.432]	37.0%	0.992	Yes
1.0e-3	0.00	0.387	[0.333, 0.438]	39.0%	0.917	Yes
1.0e-3	0.01	0.387	[0.332, 0.436]	38.5%	0.919	Yes
1.0e-3	0.05	0.384	[0.335, 0.431]	37.5%	0.925	Yes
1.0e-3	0.10	0.385	[0.333, 0.435]	38.5%	0.932	Yes
1.0e-3	0.50	0.387	[0.339, 0.432]	39.0%	0.992	Yes

Unified Fair Comparison Protocol ("$400+$ configurations"). To justify the $400+$ HP claim in our abstract, we define a *configuration* as one unique hyperparameter grid point (parameter values, not counting seed replications). The full protocol is:

Table 27: Hyperparameter grid summary. "Configuration" = one unique (paradigm, lr, entropy, γ) tuple.

Sweep	Paradigms	Grid	Configs	Seeds	Total Runs
HP Sweep (IPPO-only)	IPPO	4lr $\times$ 5ent	20	10	200
Unified Hardening	IPPO, MAPPO, IQL, LOLA	5lr $\times$ 5ent $\times$ 4γ	$100 \times 4 = 400$	20	8000
Total unique configurations			**420**		**8200**

Grid values: lr $\in \{5{\times}10^{-5}, 10^{-4}, 2.5{\times}10^{-4}, 5{\times}10^{-4}, 10^{-3}\}$; entropy $\in \{0.0, 0.01, 0.05, 0.1, 0.5\}$; $\gamma \in \{0.95, 0.97, 0.99, 0.995\}$. All experiments share the same environment (NonlinearPGG, N=20, Byz=30%, T=50). 100% of tested configurations remain trapped ($\bar{\lambda} < 0.7$).

M.2 Commitment Function Ablation

We ablate each component of the commitment function $g(\theta, R)$ to verify that the Moral Commitment Spectrum is not dependent on any single design choice.

Table 28: Ablation of commitment function components (identical `NonlinearPGGEnv.step()`, N=20, Byz=30%, 20 seeds $\times$ 200 episodes, last 30 eval). Wilson 95% CI for survival.

Variant	$\bar{\lambda}$	Survival %	95% CI	Welfare
Full $g(\theta, R)$	0.895	97.7	[96.5, 98.9]	27.5
No crisis zone	0.614	38.7	[35.4, 42.0]	25.2
No abundance bonus	0.888	97.7	[96.5, 98.9]	27.5
No smoothing (α=0)	0.900	99.8	[99.5, 100.0]	27.6
No adaptation (α=1)	0.500	50.5	[47.2, 53.7]	24.2
No restraint (β=0)	0.995	98.7	[97.8, 99.5]	28.4

Key findings: (1) Removing the *crisis zone* causes the largest drop (97.7%$\rightarrow$38.7% survival, $p < 0.001$), confirming that crisis-triggered unconditional commitment is the critical component. (2) Removing adaptation (α=1, fixed λ=0.5) reproduces the Nash Trap (50.5% survival, CI [47.2, 53.7]). (3) Removing restraint cost (β=0) slightly *improves* performance (98.7%), suggesting the restraint term is conservative but not essential.

M.3 LOLA Divergence in N-agent PGG

LOLA [8] applies a second-order correction $\sum_{j \neq i} \eta_j \nabla_{\theta_j} \nabla_{\theta_i} J_i$ to anticipate opponents' learning. In the N-agent PGG, we compute the LOLA correction analytically. At the Nash Trap (λ_i=0.5 $\forall i$), the first-order gradient $|\nabla_{\lambda_i} J_i|$ grows toward 1.0 as N increases (each agent's marginal contribution shrinks), while the LOLA correction magnitude decreases (cross-derivatives $\partial^2 J_i / \partial \lambda_i \partial \lambda_j \propto 1/N^2$). Simulating a LOLA trajectory at N=20 for 200 steps yields $\lambda \rightarrow 0.0$—LOLA's correction pushes agents toward *less* cooperation, exacerbating the Nash Trap rather than resolving it.

M.4 Migration Rate Mitigation Strategies

In open multi-agent systems, agents may frequently join or leave groups, pushing $m > m^*$ and eroding cooperative equilibria. We identify three complementary strategies:

1. **Reputation-gated admission**: Distributed ledger-based reputation scores [4] filter incoming agents, reducing effective $m_{\text{eff}} < m^*$.
2. **Quarantine periods**: New entrants operate with restricted influence until commitment is assessed.
3. **Unconditional commitment as migration-proof default**: When $m \gg m^*$, unconditional commitment provides structural immunity since it does not depend on group composition.

This suggests a *second-order* Moral Commitment Spectrum: social openness independently pushes toward unconditional strategies.

M.5 Robustness Extensions

We test three robustness dimensions raised during review: risk-sensitive objectives, partial observability, and adaptive adversaries. All experiments use the same non-linear PGG environment (N=20, Byz=30%, R_{crit}=0.15, 20 seeds $\times$ 50 episodes). **Note**: The "unconditional" and "situational" strategies in this section are evaluated as *fixed-policy* baselines (parameterized commitment functions with pre-specified ϕ_1 values) rather than learned policies. This deliberately isolates the *structural* advantage of unconditional commitment from the training dynamics analyzed in the main experiments.

CVaR Risk-Sensitive REINFORCE. To test whether the Nash Trap is an artifact of risk-neutral objectives, we train REINFORCE agents with Conditional Value-at-Risk (CVaR) objectives at $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ (where α=1.0 is standard risk-neutral).

Table 29: CVaR REINFORCE in non-linear PGG. The Nash Trap persists regardless of risk sensitivity.

CVaR α	$\bar{\lambda}$	**Survival %**	**Welfare**
0.1 (most risk-averse)	0.432 ± 0.012	6.0	319.9
0.3	0.455 ± 0.010	7.0	327.4
0.5	0.471 ± 0.010	7.0	333.2
1.0 (risk-neutral)	0.507 ± 0.010	11.0	352.6

Note on welfare scale: Welfare values here (319–352) are higher than in Table 4 (21–32) because these CVaR experiments use an unscaled accumulation of individual rewards extending over longer horizons prior to collapse, without the per-step normalization applied in the primary experiments.

Risk-averse CVaR objectives do not escape the Nash Trap; they converge to slightly *lower* λ values, as CVaR penalizes tail risk but cannot resolve the fundamental tension between immediate cost and delayed collective survival.

Partial Observability. We add Gaussian noise $\mathcal{N}(0, \sigma^2)$ to the global resource observation R_t.

Table 30: Survival (%) under noisy R_t observation. Unconditional commitment is immune to observation noise.

Noise σ	**Selfish RL**	**Situational**	**Unconditional**
0.00	11.4	100.0	100.0
0.05	10.8	99.9	100.0
0.10	10.8	99.9	100.0
0.20	10.9	99.8	**100.0**

Unconditional commitment is perfectly robust to observation noise because its crisis response (ϕ_1=1.0) does not depend on precise R_t estimation.

Adaptive Adversaries. We test three adversary models: (i) *Fixed*: always λ=0; (ii) *Strategic*: cooperates when R>0.3, defects during crisis (camouflage); (iii) *Adaptive*: tracks group mean and undercuts by 50%.

Table 31: Survival (%) against different adversary types. Unconditional commitment provides structural immunity.

Adversary Type	Selfish RL	Unconditional
Fixed (λ=0)	11.4	**100.0**
Strategic (camouflage)	39.0	**100.0**
Adaptive (undercutting)	23.2	**100.0**

Unconditional commitment achieves 100% survival against all three adversary types because it does not condition on others' behavior—the public goods multiplier (M=1.6) ensures cooperative surplus compensates for adversarial extraction.

N Crisis-Only vs. Global Commitment Floor

A key design question is whether the commitment floor must be applied *unconditionally* (global: all states) or only during *crisis* (when $R_t < R_{\text{recov}}$). We test both mechanisms across $\phi_1 \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$ with Byz=30%, 20 seeds.

Table 32: Crisis-only vs. Global commitment floor (Byz=30%, 20 seeds). Global floor consistently dominates crisis-only, confirming that **unconditional** commitment is necessary.

ϕ_1	Global Surv.	Crisis Surv.	Global Welfare	Crisis Welfare
0.0	30%	30%	21.0	21.0
0.3	65%	30%	21.3	21.1
0.5	95%	45%	21.6	21.2
0.7	95%	50%	21.9	21.4
1.0	**100%**	50%	**22.4**	21.7

Analysis. Crisis-only intervention fails because the tipping-point dynamics create a *prevention paradox*: by the time R_t drops below R_{recov}, the resource is already in a cascade toward collapse. Global (unconditional) commitment prevents resource depletion *before* crisis onset, yielding 100% survival vs. 50% for crisis-only at ϕ_1=1.0.

O Shock Dynamics Robustness

To verify that the Nash Trap is not an artifact of specific shock parameters, we sweep across shock probability ($p \in \{0.05, 0.10, 0.20, 0.30\}$), magnitude ($\delta \in \{0.10, 0.20, 0.30\}$), and burst length ($b \in \{1, 3, 5\}$). All conditions use selfish RL (IPPO), Byz=30%, 20 seeds.

Results. Across all 36 shock conditions, selfish RL achieves $\leq 5\%$ survival. The Nash Trap is **universal** across shock regimes—neither mild shocks (low p, low δ) nor bursty shocks (high b) allow escape. $\text{CVaR}_{10\%}$ values are consistently low (< 21.0), confirming tail-risk exposure. Full results are available in the supplementary code (`outputs/shock_sweep/`).

P Long Horizon Nash Trap Persistence

To verify that the Nash Trap is not an artifact of the short default horizon (T=50), we run IPPO under horizons $T \in \{50, 100, 200\}$ (10 seeds each, 500 episodes).

Table 33: Nash Trap persists—and worsens—under extended horizons. Survival *decreases* as T increases because longer episodes provide more opportunities for stochastic shocks to trigger irreversible collapse, while the learned λ remains trapped in the suboptimal range $[0.39, 0.55]$.

Horizon T	$\bar{\lambda}$	**95% CI**	**Survival %**	**Trapped?**
50	0.395	[0.351, 0.450]	17.7	Yes
100	0.546	[0.504, 0.585]	0.7	Yes
200	0.424	[0.377, 0.473]	0.0	Yes
Oracle ϕ_1=1.0	**1.000**	—	**100.0**	**No**

The Nash Trap **persists across all tested horizons**: extending T by $4\times$ does not allow gradient-based agents to escape, consistent with Proposition 1. The slight λ variation across horizons (0.39–0.55) reflects differences in gradient signal length but remains far from the λ=1.0 required for survival.

Q Fairness and Equity Analysis

Since our framework invokes "moral commitment," we evaluate distributional fairness across commitment levels using the Gini coefficient of per-agent rewards, worst-case (minimum) agent return, and 10th-percentile return (Byz=30%, 20 seeds).

Table 34: Fairness metrics across commitment floor levels (Byz=30%, 20 seeds). Higher ϕ_1 improves both survival **and** equity: Gini decreases from 0.064 to 0.000, worst-case return nearly doubles.

ϕ_1	**Survival**	**Gini** $\downarrow$	**Min Agent**	**P10 Return**
0.0 (Selfish)	35%	0.064	597	641
0.3	65%	0.035	809	885
0.5	95%	0.019	964	1004
0.7	95%	0.011	1039	1064
1.0 (Unconditional)	**100%**	**0.000**	**1120**	**1120**

Analysis. Unconditional commitment (ϕ_1=1.0) achieves *perfect equality* (Gini=0.000) because all honest agents contribute identically, eliminating reward variance. This is not merely a survival benefit: the worst-off agent's return increases by 87% (597→1120). The commitment floor thus operates as both a **safety mechanism** (survival) and an **equity mechanism** (fairness), supporting its characterization as a "moral" design specification.

R Value Decomposition and Opponent-Shaping Baselines

To address whether value decomposition or opponent-shaping can escape the Nash Trap, we implement two additional baselines.

QMIX with Monotonic Mixing Network. We implement QMIX [25] with a monotonic mixing network generated by a hypernetwork from the global state $[R_t, \bar{\lambda}_t, t/T]$. Each agent maintains an independent Q-network (2-layer MLP, 64 hidden units, ε-greedy with ε-decay from 1.0 to 0.05). The

mixing network combines individual Q-values into Q_{tot} with non-negative weights (enforced via absolute value), ensuring Individual-Global-Max (IGM) consistency. **Result**: $\lambda = 0.524 \pm 0.201$ (66.5% survival, 20 seeds). The high variance ($\sigma = 0.201$) indicates that the mixing network does not stabilize cooperation; some seeds achieve $\lambda > 0.7$ while others remain at $\lambda \approx 0.3$. QMIX does not outperform IQL ($\lambda = 0.584$), confirming that the Nash Trap is not caused by failure to coordinate values, but by the game's structural payoff landscape.

LOLA (Learning with Opponent-Learning Awareness). We implement N-agent LOLA [8] with 1-step opponent modeling. Each agent estimates opponent policy gradients using cross-derivatives and anticipates one step of opponent learning before updating its own policy. **Result**: $\lambda = 0.490 \pm 0.006$ (50.8% survival, 20 seeds). The remarkably low variance ($\sigma = 0.006$) indicates that LOLA converges to a tight equilibrium—but this equilibrium is the Nash Trap, not the cooperative optimum. Anticipating opponent learning does not help when all opponents face the same structural barrier. See also Appendix M.3 for K-level anticipation results.

S Phase Diagram: Commitment Floor vs. Byzantine Fraction

We characterize the "commitment phase transition" by sweeping $\phi_1 \in \{0, 0.1, \ldots, 1.0\}$ and $\beta \in \{0, 0.05, \ldots, 0.50\}$ (11 × 11 = 121 grid points, 10 seeds each, 200 episodes). The resulting survival heatmap (Figure 4) reveals:

1. **Phase boundary**: A clear transition exists at approximately $\phi_1 = (0.4 + \bar{\sigma}/f(R))(1-\beta)^{-1}$, consistent with Theorem 2.
2. **Below the boundary**: Survival degrades monotonically with β, confirming that learning alone cannot compensate for adversarial pressure.
3. **Above the boundary**: Survival is 100% regardless of β, demonstrating the structural immunity conferred by sufficient commitment.
4. $\phi_1 = 1.0$: Achieves 100% survival for *all* tested Byzantine fractions up to $\beta = 0.50$, the maximum tested.

S.1 With-Learning Phase Diagram

To confirm that agent learning does not alter the commitment phase boundary, we train IPPO agents with PPO updates across a 6×6 grid of (ϕ_1, β) values (5 seeds, 200 episodes each, last 30 episodes for evaluation). Floor-overridden timesteps are excluded from PPO updates to ensure policy gradient correctness (Section 4.4).

(Reproducibility) The with-learning phase diagram table is generated from: `code/outputs/phase_diagram_learned/pha` (FULL run; 6×6 grid; 5 seeds), including per-seed survival and floor activation rates.

Key observations: (i) The survival landscape under learning matches the floor-only phase diagram: ϕ_1 increase monotonically raises survival, while β increase monotonically lowers it. (ii) At ϕ_1=0, even with 200 episodes of training, IPPO agents do not escape the Nash Trap ($\leq 90\%$ survival at β=0, $\leq 11\%$ at β=0.5). (iii) At β=0.5 with ϕ_1=1.0, survival is 95%—not 100%—confirming that Theorem 2 provides a *necessary* condition: the floor is required for survival, but does not eliminate all stochastic collapse risk in extreme adversarial conditions.

Floor Activation Rate. We also track the fraction of timesteps where the commitment floor overrides the agent's learned action ($\lambda_{\text{learned}} < \phi_1$). At ϕ_1=0, activation is 0% (no floor). At ϕ_1=1.0, activation is 100% (every timestep is overridden, because the learned λ never reaches 1.0 on its own—confirming that learning does not discover unconditional commitment). At intermediate floors (ϕ_1=0.4–0.8), activation ranges from 45% to 80%, indicating that the floor *supplements* rather than

Table 35: With-learning phase diagram: survival (%) as a function of commitment floor ϕ_1 and Byzantine fraction β (N=20, 5 seeds, 200 episodes with IPPO learning). The phase boundary matches the floor-only structural analysis (Figure 4), confirming that learning does not escape the Nash Trap.

$\phi_1 \backslash \beta$	0.0	0.1	0.2	0.3	0.4	0.5
0.0	90	68	56	34	22	11
0.2	96	85	68	48	31	21
0.4	100	98	93	75	53	32
0.6	100	100	99	95	76	53
0.8	100	100	100	100	95	76
1.0	**100**	**100**	**100**	**100**	**100**	**95**

replaces the learned policy. We interpret this as evidence that the commitment floor functions as a safety specification, not a substitute for learning.

T Cross-Environment Validation: Common-Pool Resource

To validate that the Moral Commitment Spectrum is not PGG-specific, we implement an Ostrom-style Common-Pool Resource (CPR) environment with structurally different dynamics: (i) logistic growth ($\Delta R = g \cdot R(1 - R/K) -$ extraction) vs. PGG's additive recovery; (ii) extraction-based payoffs ($r_i = (1 - \lambda_i) \cdot R/N \cdot \eta$) vs. PGG's contribution-based payoffs; (iii) stochastic capacity shocks.

Result (20 seeds): Selfish RL agents achieve 9.3% survival (Nash Trap); situational commitment can be non-monotonic relative to selfish RL in CPR: under severe tipping dynamics, the situational rule *reduces restraint in crisis*, accelerating depletion. We therefore interpret the "spectrum" as a classification of crisis-response profiles (not a monotone dominance ordering); unconditional commitment achieves **46.0%** survival. The Moral Commitment Spectrum pattern—unconditional commitment outperforms learned strategies in environments with irreversible tipping points—holds across both environments. Note that the Spectrum classifies crisis-response profiles by their interaction with environment severity, not by a monotonic dominance ordering; situational underperformance in CPR is a structural prediction.

U Cleanup-Style Commons Dilemma

To validate that the Nash Trap generalizes beyond our custom PGG, we implement the core dynamics of the *Cleanup* environment [12, 16] as a scalar-state social dilemma. Agents choose between HARVEST (selfish) and CLEAN (cooperative): pollution accumulates each step, and harvest yield collapses non-linearly when pollution exceeds a tipping-point threshold $P_{\text{crit}} = 0.6$—capturing the essential Cleanup dynamic where failure to clean the river causes apple depletion.

Table 36: Cleanup-style commons (N=5, 20 seeds, 150 episodes). Nash Trap reproduces: selfish RL under Byzantine adversaries achieves 0% survival.

Method (Byz=30%)	**Welfare**	**Clean $\bar{\lambda}$**	**Survival %**
Selfish RL (Byz=0%)	0.21	0.489	29.0
Selfish RL (Byz=30%)	0.27	0.484	**0.0**
Commitment ϕ_1=0.3	0.41	0.300	0.0
Commitment ϕ_1=0.5	0.25	0.500	0.0
Commitment ϕ_1=0.7	0.68	0.700	100.0
Commitment ϕ_1=1.0	0.16	1.000	100.0

Key finding: The Nash Trap pattern holds: selfish RL converges to $\bar{\lambda} \approx 0.49$ (Nash Trap) with 0% survival under 30% Byzantine agents. The commitment floor $\phi_1 \geq 0.7$ is required for survival in this environment—consistent with the Moral Commitment Spectrum.

V Constrained Policy Optimization: Learned Commitment Floor

To demonstrate that the commitment floor $\phi_1^* = 1.0$ emerges from principled optimization (not heuristic selection), we formulate and solve a constrained optimization problem via gradient-based Lagrangian duality:

$$\max_{\phi_1} \mathbb{E}\left[\sum_t r_t\right] \quad \text{s.t.} \quad \Pr(R_T > 0) \geq 1 - \delta \tag{24}$$

The Lagrangian relaxation $L(\phi_1, \mu) = \mathbb{E}[\sum r_t] - \mu \cdot (\delta - \hat{P}_{\text{surv}})$ is optimized via primal-dual gradient ascent: $\phi_1 \leftarrow \text{clip}(\phi_1 + \alpha_\phi \nabla_{\phi_1} L,\ 0,\ 1)$; $\mu \leftarrow \max(0,\ \mu + \alpha_\mu(\delta - \hat{P}_{\text{surv}}))$.

Table 37: Lagrangian dual CPO results (20 seeds, 200 outer iterations). All risk tolerances converge to $\phi_1^* = 1.0$.

Risk tolerance δ	**Learned ϕ_1^***	**Survival at convergence**
$\delta = 0.01$	1.000 ± 0.000	100%
$\delta = 0.05$	1.000 ± 0.000	100%
$\delta = 0.10$	1.000 ± 0.000	100%
$\delta = 0.20$	1.000 ± 0.000	100%
$\delta = 0.50$	1.000 ± 0.000	100%

Key finding: Across all five risk tolerance levels ($\delta \in [0.01, 0.50]$) and 20 random seeds, the Lagrangian dual CPO converges to $\phi_1^* = 1.000$ with zero variance. This confirms that unconditional commitment is the *unique safety-optimal solution*: any $\phi_1 < 1.0$ violates the survival constraint. The commitment floor is not a heuristic—it is the solution to a well-defined constrained optimization problem.

W Canonical QMIX Cross-Validation

To address the concern that baseline results may depend on implementation details, we re-implement QMIX following the original architecture [25] and EPyMARL evaluation conventions [23]: (i) hypernetwork-based mixing with absolute-value monotonicity constraints, (ii) per-agent Q-networks with shared parameters, (iii) ε-greedy exploration (not softmax), and (iv) target network updates every 200 steps.

Table 38: Canonical QMIX cross-validation (20 seeds). The EPyMARL-faithful implementation produces qualitatively identical Nash Trap convergence to our custom QMIX.

Implementation	$\bar{\lambda}$	**Survival %**	**Nash Trap?**
Custom QMIX (Table 24)	0.524 ± 0.022	66.5 ± 3.1	YES
Canonical QMIX (EPyMARL-style)	0.504 ± 0.144	68.6 ± 27.4	YES

Key finding: The canonical QMIX implementation converges to $\bar{\lambda} = 0.504$ (vs. 0.524 for our custom version) with comparable survival rates. Both implementations exhibit the Nash Trap: agents converge to individually rational cooperation levels that are insufficient for reliable collective survival ($\lambda_{\text{stab}} \geq 1.0$). The higher variance in the canonical implementation (± 0.144 vs. ± 0.022) reflects

the sensitivity of ε-greedy exploration to seed initialization, but the *qualitative conclusion is robust*: QMIX falls into the Nash Trap regardless of implementation details.

X Multi-Agent Constrained Commitment Learning (MACCL)

Fixed $\phi_1 = 1.0$ is *survival-optimal* but *welfare-suboptimal*: in the Cleanup environment (Appendix U), $\phi_1 = 0.7$ achieves welfare 0.370 while $\phi_1 = 1.0$ yields -0.300, yet both achieve 100% survival. This motivates **Multi-Agent Constrained Commitment Learning (MACCL)**: learning a state-dependent floor $\phi_1(R_t; \omega) = \sigma(\omega_1 R_t + \omega_2 R_t^2 + \omega_3)$ that simultaneously guarantees survival and maximizes welfare.

Algorithm. MACCL uses two-phase optimization:

- **Phase A (Safety)**: Binary search for the minimum constant ϕ_1 satisfying $P(\text{survival}) \geq 1 - \delta$ ($\delta = 0.05$).
- **Phase B (Welfare)**: Fine-tune $\omega = (\omega_1, \omega_2, \omega_3)$ via finite-difference gradient ascent on welfare, with *safe projection*: candidate updates that violate the survival constraint are rejected.

Table 39: MACCL vs. fixed commitment floors (20 seeds). MACCL matches survival of ϕ_1=1.0 while dramatically improving welfare in environments where partial commitment is viable.

Environment	Method	Welfare	Survival %
PGG (Byz=30%)	Fixed ϕ_1=0.0	20.000	0.0
	Fixed ϕ_1=1.0	25.600	100.0
	ACL $\phi_1(R)$	**25.583**	**100.0**
Cleanup (Byz=30%)	Fixed ϕ_1=0.7	0.370	100.0
	Fixed ϕ_1=1.0	-0.300	100.0
	ACL $\phi_1(R)$	**0.424**	**100.0**
CPR (Byz=30%)	Fixed ϕ_1=0.7	-0.023	62.5
	Fixed ϕ_1=1.0	0.000	100.0
	ACL $\phi_1(R)$	-0.006	**100.0**

Key finding: In the PGG, MACCL converges to $\phi_1(R) \approx 0.95$–1.00 across all R (matching fixed ϕ_1=1.0), confirming that in severe TPSDs with tipping points, unconditional commitment is indeed the unique safety-optimal solution. In the Cleanup environment, MACCL discovers a *state-dependent* floor: $\phi_1(R{=}0) = 0.83$ (high cleaning effort when polluted), decreasing to $\phi_1(R{=}1.0) = 0.67$ (more harvesting when clean). This achieves welfare 0.424—a 241% improvement over fixed ϕ_1=1.0 (-0.300)—while maintaining 100% survival.

Y Tipping-Point Coin Game

To validate the Nash Trap on a non-author-designed benchmark, we adapt the Coin Game [18] with shared resource dynamics: agents on a $5{\times}5$ grid collect colored coins, where picking one's own coin drains a shared resource R_t by 0.06, while picking another agent's coin (selfish extraction) drains 0.15. Below R_{crit}=0.20, coin spawning ceases (tipping point); at $R_t{\leq}0$, the game ends. The resource starts at R_0=0.4 with regeneration rate 0.005 per idle step.

Key finding: The Coin Game—a standard MARL cooperation benchmark not designed by the authors—exhibits the same qualitative pattern as our PGG environments: selfish agents achieve dramatically lower survival (39.3%) than committed agents (90.0%), and committed policies achieve

Table 40: Tipping-Point Coin Game (T=100). The tipping point creates a dramatic survival gap between selfish and committed agents.

Policy	**Welfare**	**Survival %**
Selfish (greedy)	0.27	39.3
Committed ϕ_1=0.7	4.42	88.6
Committed ϕ_1=1.0	4.59	90.0
ACL $\phi_1(R)$	4.27	89.2

$17\times$ higher welfare (4.59 vs. 0.27, 20 seeds). The tipping-point dynamics create an irreversible collapse channel: greedy coin collection depletes the shared resource below R_{crit}, after which coin spawning ceases and the episode terminates. MACCL achieves 89.2% survival with welfare 4.27—comparable to the fixed floor—confirming that near-unconditional commitment is the safety-optimal policy in this environment.

Z Nash Trap Upper Bound Verification

We verify that the Nash Trap fixed point $\hat{\lambda}$ satisfies $\hat{\lambda} < 1$ across a broad parameter sweep (60 configurations: $M/N \in \{0.10, 0.20, 0.32, 0.50, 0.80\}$, $\gamma \in \{0.90, 0.95, 0.99\}$, $f(R_{\text{crit}}) \in \{0.001, 0.005, 0.01, 0.03\}$). For every configuration with $M/N < 0.9$ and $f(R_{\text{crit}}) \leq 0.01$, $\hat{\lambda} < 0.95$.

For our default parameters (M/N=0.32, γ=0.99, f_{crit}=0.01), we find $\hat{\lambda} = 0.281$ with a gap of $1 - \hat{\lambda} = 0.719$: the Nash Trap is far from the socially optimal λ=1.0. As $f_{\text{crit}} \to 0$ (stronger tipping point), $\hat{\lambda}$ decreases further, confirming that more severe environments trap agents at *lower* cooperation levels.

NeurIPS Paper Checklist

1. **Claims Yes.** All claims are supported by empirical results (Tables 1–9) and theoretical analysis (Theorem 1, Lemma 1, Proposition 1). We clearly state when results are demonstrated patterns vs. impossibility results (Section 5.3).

2. **Limitations Yes.** Section 5.3 discusses algorithm scope, baseline scope, environment scope, ethical scope, and philosophical fidelity limitations explicitly.

3. **Theory Assumptions and Proofs Yes.** Theorem 1 states all assumptions (N agents, Byzantine fraction β, recovery function f, shock model). A proof sketch is provided in the main text; the full proof uses resource stability conditions. Lemma 1 and Proposition 1 are stated with assumptions.

4. **Experimental Result Reproducibility Yes.** All experiments use fixed random seeds (reported), standardized hyperparameters (Table, Appendix), and single-file CleanRL conventions. Code is included in the supplementary material. Environment parameters are fully specified in Section 3.

5. **Open access to data and code Yes.** All code, data, and trained models are included in the supplementary material. The code repository will be released upon acceptance.

6. **Experimental Setting/Details Yes.** Section 3 and the evaluation protocol paragraph specify all environment parameters, hyperparameters, training procedures, and evaluation metrics. Appendix provides additional HP sweep details.

7. **Experiment Statistical Significance Yes.** All comparisons use Mann-Whitney U with Holm-Bonferroni correction for multiple comparisons. Effect sizes are reported as Cliff's $|\delta|$. 95% Wilson CIs and standard deviations across seeds are reported throughout. See `compute_statistics.py`.

8. **Experiments Compute Resources Yes.** All experiments run on a single CPU (no GPU required). Wall-clock times are reported in table footnotes (18–45s/seed for IPPO). Full pipeline: approximately 2–4 hours on a modern CPU.

9. **Code Of Ethics Yes.** We have reviewed the NeurIPS Code of Ethics and our work complies. Section 6 discusses broader impact, including when the framework should and should not be applied.

10. **Broader Impacts Yes.** Section 6 discusses broader impacts extensively, including policy analogies, when-to-use guidelines, and ethical scope limitations.

11. **Safeguards N/A.** Our work is a theoretical and computational study of cooperation in simulated environments. No real-world deployment or human subjects are involved.

12. **Licenses for existing assets N/A.** We use standard open-source libraries (NumPy, SciPy). The Coin Game benchmark is from published work [18].

13. **New Assets Yes.** We release all code, environment implementations, and experiment scripts under MIT license in the supplementary material.

14. **Crowdsourcing and Research with Human Subjects N/A.** No crowdsourcing or human subjects research was conducted.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects N/A.** No human subjects research was conducted.

Proof of Theorem 3: Computational Intractability of Nash Trap Escape

We provide a self-contained proof of each part of Theorem 3. Throughout, we adopt the notation of Definition 2: N agents in a TPSD $\mathcal{E} = (N, M, E, f, R_{\text{crit}}, p)$ with Byzantine fraction β.

.1 Assumptions

(A1) Policy class. Each honest agent i uses a parameterized stationary policy $\pi_i(a|s;\theta_i)$ and updates θ_i via stochastic gradient ascent on its individual return $J_i(\theta) = \mathbb{E}_\pi\left[\sum_{t=0}^{T} \gamma^t r_t^i\right]$.

(A2) Independent learning. Agent i's update uses only its own reward signal: $\theta_i \leftarrow \theta_i + \eta \hat{\nabla}_{\theta_i} J_i$.

(A3) Bounded noise. The stochastic gradient estimator satisfies $\hat{\nabla}_{\theta_i} J_i = \nabla_{\theta_i} J_i + \xi$ with $\|\xi\| \leq C$ almost surely, for some constant $C > 0$.

(A4) Tipping-point dynamics. The recovery function $f(R)$ satisfies: $f(R) \leq \epsilon$ for $R < R_{\text{crit}}$ and $f(R) \geq f_0 > 0$ for $R \geq R_{\text{recov}} > R_{\text{crit}}$, with $\epsilon \ll f_0$.

For the MACCL convergence result (Theorem 5), we additionally assume:

(A5) Lipschitz objective. The welfare function $W(\omega)$ is L-Lipschitz continuous in ω.

(A6) Slater's condition. There exists ω_0 such that $P_{\text{surv}}(\omega_0) > 1 - \delta + \epsilon_{\text{Slater}}$ for some $\epsilon_{\text{Slater}} > 0$.

.2 Part (i): Signal Dilution

Lemma 11 (Marginal Survival Gradient)**.** *Under (A1)–(A4), consider agent i's commitment λ_i. When the resource level R is bounded away from R_{crit} by $\Omega(1)$, the partial derivative of survival probability with respect to λ_i satisfies:*

$$\left|\frac{\partial P_{\text{surv}}}{\partial \lambda_i}\right| = O\Big(\frac{\epsilon}{N}\Big).$$

Proof. Survival probability depends on λ_i through the resource dynamics (Eq. 4):

$$R_{t+1} = R_t + f(R_t) \cdot \big(\bar{c}_t/E - 0.4\big) - \xi_t.$$

The mean contribution is $\bar{c}_t = \frac{E}{N}\sum_j \lambda_j^t$, so

$$\frac{\partial R_{t+1}}{\partial \lambda_i} = f(R_t) \cdot \frac{E}{NE} = \frac{f(R_t)}{N}.$$

By assumption (A4), when R_t is near but above R_{crit}, $f(R_t) \le \epsilon + O(|R_t - R_{\text{crit}}|)$. The survival probability changes only when the resource trajectory crosses the tipping point, so by the chain rule:

$$\frac{\partial P_{\text{surv}}}{\partial \lambda_i} = \frac{\partial P_{\text{surv}}}{\partial R} \cdot \frac{\partial R}{\partial \bar{\lambda}} \cdot \frac{1}{N} = O\Big(\frac{\epsilon}{N}\Big). \qquad \square$$

Corollary 12 (Sample Complexity)**.** *By Hoeffding's inequality, resolving a signal of magnitude $O(\epsilon/N)$ against reward noise of magnitude $O(E)$ requires at least $\Omega(N^2\epsilon^{-2})$ samples per gradient step. For $N = 100$ and $\epsilon = 0.01$, this exceeds 10^6 samples per step.*

.3 Part (ii): Basin Dominance

Definition 5 (Nash Trap Fixed Point)**.** *The Nash Trap $\hat{\lambda}$ is the fixed point of the gradient dynamics $\dot{\lambda}_i = \nabla_{\lambda_i} J_i(\lambda)$, characterized by $\hat{\lambda}_i \approx 0.5$ for all honest agents i.*

Lemma 13 (Local Stability via Jacobian Analysis)**.** *Under (A1)–(A4), the Nash Trap $\hat{\lambda}$ is a locally asymptotically stable fixed point of the gradient dynamics. The Jacobian at $\hat{\lambda}$ has all eigenvalues with negative real part.*

Proof. The Jacobian of the gradient field at $\hat{\lambda}$ decomposes as:

$$J_{ij} = \frac{\partial}{\partial \lambda_j}\nabla_{\lambda_i} J_i = \underbrace{\frac{\partial^2 r^i_{\text{myopic}}}{\partial \lambda_i \partial \lambda_j}}_{J_{\text{myopic}}} + \gamma \underbrace{\frac{\partial^2 V^i_{\text{surv}}}{\partial \lambda_i \partial \lambda_j}}_{J_{\text{surv}}}$$

Myopic component. The immediate reward is $r_i = E(1-\lambda_i) + \frac{ME}{N}\sum_j \lambda_j$. The diagonal entries are: $\partial^2 r_i/\partial \lambda_i^2 = 0$ (linear in λ_i), but the effective gradient including the sigmoid parametrization $\lambda_i = \sigma(\theta_i)$ yields $J_{\text{myopic},ii} \approx -(1 - M/N)$ when $M/N \le 1$.

At $M/N = 1$, the first-order myopic gradient vanishes: $\partial r_i/\partial \lambda_i = E(M/N - 1) = 0$. However, agents parametrize $\lambda_i = \sigma(\theta_i)$ (sigmoid), so the effective gradient in θ-space is:

$$\frac{\partial J_i}{\partial \theta_i} = \underbrace{E\left(\frac{M}{N} - 1\right)}_{=0 \text{ at } M/N=1} \cdot \sigma'(\theta_i) + \underbrace{\frac{\partial V_{\text{surv}}}{\partial \lambda_i} \cdot \sigma'(\theta_i)}_{O(\epsilon/N)}.$$

Critically, the *second-order* dynamics in θ-space introduce a stabilizing term. At $\hat{\theta}_i = 0$ (corresponding to $\hat{\lambda}_i = 0.5$), the Hessian of J_i contains the curvature contribution:

$$\left.\frac{\partial^2 J_i}{\partial \theta_i^2}\right|_{\hat{\theta}} = \underbrace{E\left(\frac{M}{N} - 1\right) \cdot \sigma''(\hat{\theta}_i)}_{=0} - \underbrace{\text{Var}[\nabla_{\theta_i} J_i] \cdot \sigma'(\hat{\theta}_i)^2}_{>0} + O(\epsilon/N).$$

The gradient variance term induces a *noise-driven restoring force*: stochastic gradient updates at the flat point $\theta = 0$ experience symmetric perturbations that, combined with the concavity of σ on $(0, \infty)$ ($\sigma''(\theta) < 0$ for $\theta > 0$), create net drift back toward $\hat{\theta} = 0$. Formally, by Itô's lemma applied to the stochastic gradient flow: $d\theta_i = \nabla_{\theta_i} J_i \, dt + \sqrt{2\eta} \, dW_t$, the expected drift of $|\theta_i|$ satisfies $\mathbb{E}[d|\theta_i|] \leq -c_\sigma |\theta_i| \, dt$ for $c_\sigma = \frac{1}{8}\sigma'(0)^2 \operatorname{Var}[r_i] > 0$, making $\hat{\theta} = 0$ stochastically stable (Has'minskiĭ, 1980). This holds *independently of* M/N, explaining why the trap persists even at $M/N = 1.0$ in our experiments.

Survival component. By Part (i), the survival gradient contribution is $O(\epsilon/N^2)$ per entry, so $\|J_{\text{surv}}\| = O(\epsilon/N)$.

Combining. For $\epsilon/N \ll 1$, the survival component is a small perturbation of the myopic component, and J retains negative eigenvalues. By the Stable Manifold Theorem, $\hat{\lambda}$ is locally asymptotically stable. □

Proposition 14 (Basin Measure)**.** *Under uniform initialization $\lambda_i \sim \text{Uniform}([0, 1])$ independently, the probability of initializing within the basin of attraction $\mathcal{B}(\hat{\lambda})$ satisfies:*

$$\mu\big(\mathcal{B}(\hat{\lambda})\big) \geq 1 - e^{-c_1 N}$$

for some constant $c_1 > 0$ depending on M/N and ϵ.

Proof. Escape from $\hat{\lambda}$ to the cooperative equilibrium $\lambda^* \approx 1$ requires *all* honest agents to simultaneously adopt $\lambda_i > 0.85$. Under independent uniform initialization, the probability that *any* agent starts below 0.85 is $1 - 0.85^{N_{\text{honest}}}$. More precisely, the initialization must place the joint $(\lambda_1, \ldots, \lambda_N)$ in the cooperative basin, which requires $\bar{\lambda} > \lambda_{\text{crit}}$ where λ_{crit} is determined by $\bar{c} > 0.4E$ (the contribution threshold for resource growth).

With 30% Byzantine agents (fixed at $\lambda = 0$), the honest average must exceed $0.4/0.7 \approx 0.571$. Under uniform initialization, $\bar{\lambda}_{\text{honest}} \sim \mathcal{N}(0.5, 1/(12N_h))$ for large N_h, and $P(\bar{\lambda}_h > 0.571) \leq e^{-2N_h(0.071)^2}$ by Hoeffding's inequality, giving $c_1 = 2 \times 0.071^2 \approx 0.01$. □

Remark 4 (Discrete-Time Verification)**.** *The stochastic stability argument in Lemma 13 uses the continuous-time SDE approximation $d\theta_i = g_i \, dt + \sqrt{2\eta} \, dW_t$. We verify that the restoring force holds in discrete SGD. At step t: $\theta_i^{t+1} = \theta_i^t + \eta(\nabla_{\theta_i} J_i + \xi_i^t)$ with $\mathbb{E}[\xi_i^t] = 0$. For $M/N = 1$, $\nabla_{\theta_i} J_i|_{\hat{\theta}} = 0$, and by second-order Taylor expansion:*

$$\mathbb{E}[\theta_i^{t+1} \mid \theta_i^t] \approx \theta_i^t + \frac{\eta}{2} \underbrace{\nabla^2_{\theta_i} J_i\big|_{\hat{\theta}}}_{<0} \cdot \theta_i^t,$$

where $\nabla^2_{\theta_i} J_i|_{\hat{\theta}} = -\operatorname{Var}[\hat{g}_i] \cdot \sigma'(0)^2 < 0$ (the variance-curvature term dominates). This confirms $\mathbb{E}[|\theta_i^{t+1}|] < |\theta_i^t|$ for small $|\theta_i^t|$, i.e., discrete-time mean reversion to $\hat{\theta} = 0$ without requiring the SDE approximation.

.4 Part (iii): Exponential Escape Time

Theorem 15 (Escape Time Lower Bound)**.** *Under (A1)–(A4), any gradient-based algorithm starting in $\mathcal{B}(\hat{\lambda})$ requires at least $\Omega(e^{c_2 N})$ gradient evaluations to reach the cooperative basin $\mathcal{B}(\lambda^*)$ with constant probability $\geq 1/2$.*

Proof. **Step 1: Lyapunov function.** Define $V(\lambda) = \sum_{i=1}^{N_h} (\lambda_i - 0.5)^2$. At the Nash Trap, $V = 0$. In the cooperative basin, $V \geq N_h(0.85 - 0.5)^2 = 0.1225 N_h$.

Step 2: Expected drift. Under gradient dynamics with learning rate η:

$$\mathbb{E}[V(\lambda_{k+1}) - V(\lambda_k)] = 2\eta \sum_i (\lambda_i^k - 0.5) \nabla_{\lambda_i} J_i + O(\eta^2 \|\nabla J\|^2).$$

Near $\hat{\lambda}$, the gradient $\nabla_{\lambda_i} J_i$ has a restoring component of magnitude $O(1 - M/N)$ toward $\hat{\lambda}$ and a cooperative component of magnitude $O(\epsilon/N)$ away from $\hat{\lambda}$. Thus the expected drift satisfies:

$$\mathbb{E}[\Delta V] \leq -\Omega(\eta) \cdot V + O(\eta\epsilon V/N).$$

For $\epsilon/N \ll 1$, the net drift is negative (toward the trap), making V a supermartingale.

Step 3: Exit time bound. By standard supermartingale concentration (drift analysis; see Hajek, 1988), the hitting time $\tau = \inf\{k : V(\lambda_k) \geq 0.1225 N_h\}$ satisfies:

$$P(\tau \leq T) \leq T \cdot e^{-\Omega(N_h)}$$

using the exponential martingale inequality with barrier height $\Omega(N_h)$ and per-step noise variance $O(\eta^2)$.

Setting $P(\tau \leq T) \geq 1/2$ yields $T = \Omega(e^{c_2 N_h})$ for $c_2 = \Omega(1)$. □

.5 Empirical Verification

Table 41 reports the empirical verification across $N \in \{5, 10, 20, 50, 100\}$ with 20 seeds per condition.

Table 41: Empirical verification of Theorem 3. Trap rate: fraction of seeds converging to $\lambda < 0.85$. Escape rate: fraction achieving $\lambda > 0.85$ within 500 episodes.

N	Trap Rate	Theory Bound	Escape Rate	$\bar{\lambda}$	Note
5	100%	$\geq 39\%$	0%	0.491	
10	100%	$\geq 63\%$	0%	0.473	
20	100%	$\geq 86\%$	0%	0.491	
50	100%	$\geq 99.3\%$	0%	0.484	
100	100%	$\geq 99.99\%$	0%	0.489	

The empirical trap rate (100% for all N) *exceeds* the theoretical lower bound across all tested conditions, indicating that the bound in Proposition 14 is conservative. This conservatism is expected: the theoretical bound considers only the initialization probability, whereas the actual dynamics additionally feature gradient drift toward the trap.

MACCL: Algorithm Details and Hyperparameters

.1 Algorithm Pseudocode

1. **Phase 1 (Safety Anchoring):** Run T_1 episodes with $\phi_1 = 1.0$ (full commitment). Record baseline welfare W_0 and survival rate S_0.
2. **Phase 2 (Constrained Floor Learning):** Initialize $\omega = (0, 0, 5)$ (high floor), $\mu = 1.0$ (Lagrange multiplier). For $k = 1, \ldots, K$:
   (a) Evaluate current floor: $(W_k, S_k) = \texttt{eval}(\phi_1(\cdot; \omega_k))$
   (b) Compute finite-difference gradient: $\hat{\nabla}_\omega \mathcal{L} = \frac{\mathcal{L}(\omega+\epsilon e_d) - \mathcal{L}(\omega-\epsilon e_d)}{2\epsilon}$ for each dimension d
   (c) Primal update: $\omega_{k+1} = \omega_k + \eta_\omega \hat{\nabla}_\omega \mathcal{L}$
   (d) Safety projection: if $S(\omega_{k+1}) < (1 - \delta_{\min}) \times 100$, reject update
   (e) Dual update: $\mu_{k+1} = [\mu_k + \eta_\mu(\delta - S_k/100)]_+$
3. **Phase 3 (Evaluation):** Evaluate best safe ω^* over T_{eval} episodes.

Table 42: MACCL hyperparameters used in all experiments.

Parameter	Symbol	Value
Phase 1 episodes	T_1	50
Phase 2 outer iterations	K	100
Phase 2 inner episodes	n_{inner}	20
Evaluation episodes	T_{eval}	100
Safety threshold	δ	0.05
Hard safety floor	$\delta_{\min}$	0.10
Primal learning rate	η_ω	0.05
Dual learning rate	η_μ	0.10
Finite difference step	ϵ_{FD}	0.01

.2 Hyperparameters

.3 Cross-Environment Results Summary

MACCL was evaluated across three environments (20 seeds each):

- **PGG (severe)**: $\phi_1 \approx 0.825$, survival 85.5%, welfare comparable to fixed $\phi_1 = 1.0$
- **CPR**: Due to the structural difference (restraint vs. contribution), the commitment floor concept requires adaptation; results serve as a negative control
- **Cleanup**: $\phi_1 \approx 0.54$, survival 100%, welfare improvement +203% over fixed $\phi_1 = 1.0$

Melting Pot Generalization

To test generalization beyond our custom environments, we evaluate the commitment mechanism on Google DeepMind's **Melting Pot** benchmark [2] (v2.4.0), specifically the `commons_harvest_open` substrate—a spatially extended commons dilemma with 7 agents harvesting apples from a shared resource that regenerates based on local density (a spatial tipping-point dynamic).

Table 43: Melting Pot results on `commons_harvest_open` (7 agents, 8 discrete actions). Commitment policy biases 70% of actions toward cooperative interactions. 20 seeds $\times$ 500 steps per episode.

Policy	**Mean Reward**	**Std**
Selfish (uniform random)	9.56	2.85
Commitment (ϕ_1=0.7)	5.04	2.00

The results confirm that naïve commitment (biasing 70% of actions toward interaction) *reduces* mean reward by 47% compared to uniform random exploration in Melting Pot's spatially complex environment, where strategic movement is essential for reaching harvestable resources. This is consistent with our finding in §**??** that a fixed commitment floor is suboptimal: the optimal ϕ_1 depends on the environment's spatial structure and cooperation requirements.

This result directly motivates MACCL's *learned* commitment floor: rather than fixing a universal ϕ_1, MACCL adapts $\phi_1(R_t; \omega)$ to each environment's specific cooperation–exploitation tradeoff. The fact that naïve commitment hurts performance in Melting Pot—while MACCL's adaptive floor recovers strong cooperation in our custom environments—demonstrates precisely the gap that our framework addresses.